April 20, 2017

DIAMOND IMPROVEMENT DISTRICT

Diamond Improvement District

Water System Evaluation



McElhanney

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April 20, 2017

Diamond Improvement District
Box 747
Ladysmith, BC
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Attention: Mr. Larry Williams, Chairman

Re: Water System Evaluation

Dear Mr. Williams,

Please find a final copy of the Water System Evaluation for the Diamond Improvement District. This document has been revised per comments received in your email dated March 8, 2017 and per comments received by phone on April 6, 2017.

Our report recommends replacing much of the district's watermains. There are two primary concerns;

1) 100mm Diameter Watermains – These small diameter watermains cannot supply fire flows as required by the Letters Patent, and
2) Asbestos Cement (AC) Watermains – AC watermains are at the end their expected service life. Local governments are commonly engaged AC replacement program due to prevalence of breaks and leaks.

Please review the document and advise if you have any comments or questions and/or if would like to make any revisions.

Yours truly,
McELHANNEY CONSULTING SERVICES LTD.



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1.1. Introduction

McElhanney Consulting Services Ltd (McElhanney) was retained by the Diamond Improvement District (Diamond) to complete a Water System Evaluation study to summarize the condition of the existing water system and plan for future improvements.

The tasks for this study include:

1) Topographic Survey,
2) Drawing of Existing Water System,
3) Drawing with Watermain Upgrades,
4) Class C Cost Estimate,
5) Development Potential Discussion,
6) Funding Options, and
7) Submission of this report.

The results of this Water System Evaluation study are summarized in the following sections of this report.

1.2. Background

The Diamond Improvement District is located directly north of the Town of Ladysmith, BC. This unincorporated community lies within the Cowichan Valley Reginal District and forms part of Electoral Area "H". The district borders the Stz'uminus First Nation to the North and the Town of Ladysmith to the south. The water system serves a population of approximately 530 residents.

The Diamond Improvement District Waterworks District was incorporated under the Water Act in 1953 and is an Improvement District as defined under the Local Government Act. The District's Letters patent was created in April 21, 1953 and states: "the objects of the improvement district shall be the provision of fire protection and the acquisition, maintenances, and operation of works buildings, and equipment for that purpose." The Proclamation, Letter Patent, and modifications are included in Appendix A for reference.

1.3. Reference Standard

The referenced standard for this report is the Master Municipal Construction Documents Association (MMCD) Design Guidelines 2014. "This Municipal Infrastructure Design Guidelines 2014 was prepared under the auspices of the Master Municipal Construction Document Association (MMCDA), which is an association of British Columbia Municipalities, Regional Districts, Contractors and Consultants. The purpose of the manual is to provide a standardized set of guidelines that can be adopted by Municipalities and other agencies involved with design and construction of municipal infrastructure."[1]

1.4. Reference Reports

The following is a list of known studies referenced in this report:

- The Diamond Improvement District, Fire Hydrant-Annual Inspection Report (2013-2014)
- Ladysmith Water Study, Koers & Associates Engineering Ltd, 1995
- The Town of Ladysmith Annual Water Report 2013

A copy of the Inspection Reports is included in Appendix B.

[1] Master Municipal Construction Documents Association, Design Guideline 2014

2. Existing System

2.1. Water Supply

The Diamond Improvement District receives water from the Town of Ladysmith through a Water Services Agreement. The Town of Ladysmith water system has two sources: Stocking Lake and Holland Lake. Water from both sources is piped directly to the enclosed Arbutus Reservoir where it is chlorinated and distributed by gravity to the entire Ladysmith service area. Diamond is connected to the Ladysmith water system at two points:

1. Rock Creek Road at Gladden Road, and
2. Christie Road at 3rd Avenue.

Diamond's water distribution network consists of watermains, hydrants, valves, and services. Watermains are composed primarily of asbestos-cement (AC) pipe with some polyvinyl chlorine (PVC) pipe. The existing water system is illustrated on Figure 1, appended to this report.

2.2.Water Services Agreement

A summary of the 2016 Water Services Agreement with the Town of Ladysmith is presented in Table 2-1 below.

Table 2-1 – Water Services Agreement Summary

Item	Agreement	Reference
Water Supply	70,000 cubic meters per year (2016 to 2021)	2.1
Rate	$0.62 per cubic meter of water due on a quarterly basis.	3.1, 3.3
Parcel Tax Equivalent	80% Parcel Tax Equivalent payable by Diamond to Ladysmith due July 1st of each year of Term.	4.2
DCC Costs Equivalent	Diamond shall pay Ladysmith equivalent Development Cost Charges for each new parcel or dwelling connected to water system.	5.0
Diamond Costs	Diamond responsible to pay: • for all fees related to new connections and upgrades of associated Ladysmith infrastructure, • for the installation, operation, maintenance, and, repair, and replacement of bulk water meters • construction, repair, replacement, and maintenance of the Diamond Water System.	9.7 10.2 2.5
Diamond Responsibilities	Diamond is responsible to ensure that water distributed within the Diamond Water System meets BC drinking water guidelines.	6.3

2.3. Water System Configuration

The District's water distribution system has two main loops and several linear sections with dead-ends. Looped watermains provide system redundancy and reduce headloss associated with long dead-end lines. The Diamond water distribution system benefits from having two connections to the Ladysmith water system. This provides added backup and security if one supply connection fails.

2.4. Pressure Distribution

System pressure is controlled by pressure reducing valves (PRVs) located at the two points of connection to the Town of Ladysmith distribution network. Pressure varies throughout the system in proportion to ground elevation. Highest pressures are at the bottom of the system while the lowest pressures are at the top end of the system.

The Diamond Improvement District performs annual hydrant inspections. A copy of the two previous inspection reports in included in Appendix B for reference. The hydrant tests include pressure readings. A summary of those readings is provided in Figure 2. The minimum and maximum pressures recorded are presented in Table 2-2 on the following page along with the applicable MMCD standards.

Table 2-2 – Watermain Pressure Summary with MMCD Standards

Parameter	Pressure	Notes
Minimum Recorded Pressure in Diamond Improvement District	421 kPa (61 psi)	Located at the upper end of the system on Grouhel Road.
MMCD Minimum Allowable Pressure	*300 kPa (44 psi)* *150 kPa (22 psi)*	*Minimum Pressure at Peak Hour Demand* *Minimum Pressure at Fire Flow + Maximum Day Demand*
Maximum Recorded Pressure in Diamond Improvement District	841 kPa (122psi)	Located at the low end of the system on the highway north of Jones Road.
MMCD Maximum Allowable Pressure	*850 kPa (123 psi)*	*Subject to approval of the local authority, the maximum allowable pressure may be increased to 1035 kPa (150 psi) for systems with multiple pressure zones.* *Where the maximum pressure exceeds 515 kPa (75 psi), service connections must be individually protected by pressure reducing valves located in the buildings being served.*

The recorded pressures indicate that the Diamond's water system is operating within MMCD standard allowable pressures. As noted in the above table, individual pressure reducing valves (PRVs) are required in areas with a pressure greater than 515 kPa (75 psi). Accordingly, individual PRVs are required across the Diamond Improvement District except for Grouhel Road.

2.5. Watermains

Watermain sizes range from 100mm (4") to 150mm (6"). Existing watermains are undersized by current standards. A summary of the current watermain lengths is presented in Table 2-3 below:

Table 2-3 – Watermain Summary

Pipe Type	Diameter	Total Length	Percentage of Total Length
AC	100mm	2,429m	33%
AC	150mm	2,275m	31%
PVC	150mm	2,359m	35%
PVC	200mm	70m	1%
TOTALS		**7313m**	**100%**

Minimum Pipe Sizes

The MMCD Guidelines specify a minimum pipe diameter of 200mm. This minimum size can be reduced to 150mm in residential areas for lengths less than 500m and in looped systems where fire flow requirements can be met.

2.6. Asbestos Cement Pipe

Asbestos Cement pipe was in common use prior to the introduction of Polyvinyl Chloride in the mid 1970's. Since that time many communities have experienced problems with deterioration and the eventual failure of their AC pipe system. These problems result from a combination of the following factors:

- Failures are most prevalent in districts that use a surface water supply that tends to be slightly more acidic than ground water. This 'soft' water attacks the cement and reduces pipe strength.
- AC pipe is a brittle material that is prone to cracking when subjected to uneven loads caused by trench settlement, slope movement or vehicle loads.
- Insufficient bedding during installation may result in point loads against the outside of the pipe that result in high stress and pipe failure.

Many communities in BC have adopted an AC pipe replacement program to spread the cost of upgrades over time. The warrant for action is assessed on a case by case basis depending on factors specific to the community.

2.7. Hydrant Spacing

The MMCD standard hydrant spacing is 150 m and no more that 90 m from a building. There are a total of 28 hydrants in the Diamond and spacing ranges considerably. The hydrant spacing along the newer watermain on Rocky Creek Road is less than the remaining areas in the district. A summary of hydrant spacing is provided in Table 2-4 below:

Table 2-4 – Hydrant Spacing

Area	Average Spacing (Approximate)
Entire District Area	200m
Rocky Creek Road	140m
Entire District *Less* Rocky Creek Road	230m

Hydrants along Rocky Creek Road are to current MMCD standards while the remaining areas of the district are greater than the 150m standard.

2.8. Dead-Ends

Per MMCD Standards, watermains are to be looped wherever possible. For dead-end systems, blow-offs are required. Hydrants may serve as blow-offs where approved by local authority.

3. Water Consumption

3.1. Population

The District indicated that they service 241 households. At an assumed density of 2.2 persons per household, the 2016 serviced population for the District is estimated to be 530 persons.

3.2. Historical Water Use

Historical water use records were provided by Diamond. A summary of the historical water use, population, and per capita consumption is provided in Table 3-2 below.

Table 3-1 – Historical Water Use

Year	Annual Water Use Cubic Metres (m^3)
2011	57,051
2012	66,996
2013	57,238
2014	51,671
2015	49,330
2016	47,042

Note that the total water use in 2016 was 63,643m^3 but 16,601m^3 was subtracted from this total to account for an 'extraordinary break.' The current Water Services Agreement with the Town of Ladysmith is for 70,000 m^3/year and the historical water use is within this volume. Future demands may change and could impact the Agreement.

3.3. Per Capita Consumption

The per capita use is estimated based on 2016 estimated service population and water use or:

47,042 m^3 ÷ 530 = 88.8 m3/year,

or 243 liters per person per day (L/c/d).

Table 3-3 provides a comparison of the District's water system demand rates to a range of standards from other municipal water systems on Vancouver Island.

Table 3-2 – Water Demand Comparison

Municipality or Standard	Average Day Demand L/c/d
Diamond Improvement District	**243**
Town of Ladysmith	446
City of Nanaimo	455
Capital Regional District	545
Master Municipal Construction Documents (MMCD)	600

Water demand in the Diamond Improvement District is *substantially less* than other comparable systems indicating the system integrity is good and/or per capita consumption is low.

3.4.Water-Meter Based Payments

Please note, water meters are installed on all services throughout the Diamond water system. Prior to June 2016, Diamond water users payed a bulk fee for the water consumed. Starting in June 2016, the payment changed to be based on the individual water meter readings. Water meter-based payments typically decreases consumption by at least 25%[2].

3.5. Unaccounted Water

The Diamond Improvement District advised McElhanney that bulk meter readings (water supply from Ladysmith) has been compared to the total water consumed per individual meters. This analysis indicated that the volume of water consumed is within 10% of the bulk meter readings. This is comparable to new and well managed systems. The amount of water that is lost or unaccounted for can be as high as 50% in older and poorly managed systems[3].

The Diamond Improvement District also records the consumption at both bulk meters on a weekly basis. This provides a good check to noted if a significant break has occurred within the system.

[2] Corbitt, R.A., Standard Handbook of Environmental Engineering, McGraw-Hill Handbooks, 1999.
[3] Hunaidi,O., Detecting Leaks in Water-Distribution Pipes, Construction Technology Update No.40, National Research Council's Institute for Research in Construction, Canada, 2000

3.6. Zoning and Full Build-Out

Future water consumption was estimated based on full build-out of the land areas served by the Diamond Improvement District as defined by current zoning bylaws. The total number of services was based on an assumed minimum lot size of 0.5 hectares. Actual minimum lot size will vary with sanitary sewer system requirements as regulated by the Vancouver Island Health Authority. A summary of the CVRD OCP Designation and Zoning in the Diamond Improvement District is presented in Table 3-4 below.

Table 3-3 – Zoning and Maximum Dwelling Estimate

OCP Designation	Zoning	Location	No. of Existing Lots[1]	Min. Parcel Size (Zoning)	Min. Parcel Size (Assumed)	Max. No of Lots[1]	Max. No. of Dwellings
Suburban Residential	Suburban Residential R-2	Majority of the district.	170	0.4 ha (1.0 ac)	**0.5 ha** (1.2 ac)	**211**	**211**
Suburban Residential	Institutional P-2	Diamond office, Lot south of Diamond office, and Ladysmith Little Theatre.	3	1675 m² (0.4 ac)	1675 m² (0.4 ac)	3	-
Mobile Home Park	Manufactured Home Residence MH-1	Christie Rd near Arroyo Rd and Christie Rd at Grouhel Rd.	23[2]	325 m² (0.1 ac)	325 m² (0.1 ac)	**40**	**40**
Forestry	F-2	South of Grouhel Rd, western extents of Diamond.	1	80 ha (198 ac)	80 ha (198 ac)	1	-
Agriculture	P-1	South of Grouhel Rd, eastern extents of Diamond.	1	-	-	1	-
Commercial	P-2	5091 Malamos Rd	2	1675m² (0.4 ac)	1675m² (0.4 ac)	2	-
Commercial	W-5	5025 Limberis Rd.	1	-	-	1	-

[1] - *Estimated.*
[2] - *1 lot undeveloped.*
[3] - *For Parcels Served by Community Water System Only.*

Maximum Population and Build-Out Timeframe

In summary, it is estimated that the Diamond Improvement District can grow by 58 dwellings. At an average of 2.2 persons/dwelling that is 127 people for a maximum population of 657 people. Assuming 2% annual growth rate, the full buildout could occur in 11 years (2027). At the current average consumption rate of 243 L/c/day, the full buildout demand will be 58,000 m^3/year which is less than the Water Services Agreement of 70,000 m^3/year.

3.7. Fire Flows

The District's Letters Patent include "the provision of fire protection." The British Columbia Building Code references the following documents for fire flow calculations:

- "Needed Fire Flow Guide," Insurance Services Office
- "NFPA 1142 Standard for Water Supplies for Rural and Suburban Fire Fighting," National Fire Protection Association, 2017
- "Distribution Requirements for Fire Protection," American Water Works Association (AWWA)

Required fire flows vary with building size, type of construction, and type of occupancy, and exposure to nearby hazards (other buildings within 15m). A summary of the applicable fire flow standards is presented in Table 3-5 below.

Table 3-4 – MMCD Fire Flow Standards

MMCD Standard	Value
Minimum Fire Flow	60 L/s
Velocity	3.5 m/s
Pressure	150 kPa (22 psi)

Estimated Minimum Fire Flow Demand

The estimated minimum fire flow demand for Diamond is calculated as follows:

- Average Day Demand (ADD): 243 L/c/day x 530 people = 128,900 L/day or 90 L/min
- Maximum Day Demand (MDD): 2 (assumed factor) x ADD = 2 x 90 L/min = 180 L/min
- Minimum Fire Flow (MFF): 60 L/s or 3,600 L/min
- **Minimum Fire Flow Demand =** **Fire Flow + Domestic Consumption**
  = 3,600 L/min + 180 L/min = 3,780 L/min

Pressure Losses at Minimum Fire Flow Demand

At high flows, pressure loss increases as pipe size decreases. A summary of the relative headloss at the Fire Flow Demand for a section of 100m pipe is summarized in the Table 3-6 below:

Table 3-5 – Headloss at Fire Flow Demand for 100m Pipe

Diameter mm	Headloss kPa (psi)	Velocity m/s
100	5.4 kPa (61psi)	8.0
150	0.71 kPa (8.0 psi)	3.6
200	0.17 kPa (2.0 psi)	2.0

The above table illustrates the sensitivity of pressure loss to pipe size and the need for watermains to be larger than 100mm diameter. Take, for example, the section of 477m length of 100mm AC pipe on Christie Road between the looped highway crossing and the western extents near Arroyo Road. The headloss for the fire flow demand is estimated to be 477m/100m x 61 psi = 291 psi. The static pressure per the hydrant flow tests is approximately 96 psi. Therefore, there is not enough pressure in the system to supply the headloss required for the fire flow. The pipe size should be increased to a minimum 150mm diameter which would occur a headloss of 477m/100m x 8.0psi = 38 psi. The reduction in pressure for this system would be 96 psi less 38 psi or 39 psi which is substantially above the minimum 20 psi requirement.

3.8. Water Supply & Storage

The Diamond water supply and reservoir storage requirements are provided by the Town of Ladysmith. An analysis of their ability to meet the supply, potability, and storage demands was not addressed in this report.

3.9. System Condition & Losses

There is currently no program or data for the Diamond Improvement District to monitor the condition of the watermain system or losses.

4. Water System Concerns

A summary of the water system concerns is presented in Table 4-1 below:

Table 4-1 – Water System Concerns

Item	Description
Watermain Size	• The standard minimum pipe size is 200mm, (can be reduced to 150mm in conditions where fire flows can be met). • Most Diamond's watermains are 150mm or less. • 33% of the watermains are 100mm. • 100mm watermains significantly restrict fire flows.
Watermain Age & Materials	• The improvement's watermains are primarily constructed out of asbestos-cement. • AC watermains are at the end of the expected service life. • Increasingly frequent breaks may occur in AC pipe.
Hydrant Spacing	• The standard minimum hydrant spacing is 150m. • Outside of the Rocky Creek Road watermain, average hydrant spacing is 230m.
Pressure	• System pressures are generally high but within MMCD standards. • Individual service connections require pressure reducing valves in all areas of the district except for Grouhel Road. • High pressures generally lead to high consumption.
System Looping	• There are many dead-end lines in the water system. • Watermains could be looped to improve fire flows.

5. Conclusions

A summary of conclusions is presented below:

1) 61% or 4,704 m of the district's watermain are composed of asbestos-cement pipe.
2) 100mm diameter pipe is smaller than current standards and cannot convey required fire flows.
3) Residual pressures range from 421 kPa (61 psi) to 800 kPa (116 psi) and may vary substantially.
4) Individual pressure reducing valves are required in all areas with a residual pressure greater than 515 kPa (75 psi). This applies to all areas of the district except Grouhel Road.
5) The system currently serves approximately 530 people and is projected to grow to a maximum build-out population of 657.
6) A watermain break in 2016 resulted in the loss of 16,601m^3 of water.
7) Average daily water consumption is 243 litres per capita per day. This consumption is lower than other jurisdictions;
8) Water meter-based billings began in June 2016 and per capita consumption is anticipated to be even lower in the future.
9) There is less than 10% water loss in the system.
10) Maximum build-out demand is estimated to be 58,000 m^3 per year.
11) System condition and losses are undefined. It is not known if the condition AC pipe is adequate or if there are any leaks in the district system.

6. Recommendations

Based on our review of the existing information provided, we recommend the following system improvements. Each project has been assigned a schedule of priority focused on strengthening the system to improve domestic water supply and improve residential fire flows.

Project 1 – Water Network Flow Assessment & Modeling (Priority 1-5 years)

It is recommended to complete a water network flow assessment and model for the Diamond Improvement District to:

- Assess the system for leaks and identify replacement priorities.
- Identify recommended pipe size improvements
- Model and size watermain looping scenarios.
- Prioritize replacement projects.
- Model the addition of pressure reduction scenarios (i.e. PRV installations).

Project 2 – 100mm Diameter Watermain Replacement (Priority 1-5 years)

Replacing the undersized 100mm watermains is an essential upgrade required to meet fire flow demands. The most critical pipes are at the highest elevation with the lowest residual pressures. These should be replaced first followed by the more down-gradient watermains. Watermain improvements include increasing the number of fire hydrants to meet current standards. Project 1 is required to be completed first in order to size the watermains and define where appurtenances, such as pressure-reducing valves, may be required.

Project 3 – 150mm AC Condition Assessment (Priority 1-5 years)

AC watermains are approaching the end their expected service life. All systems deteriorate with age and use; however, performance may not be affected until deterioration is well advanced. The frequency of watermain breaks like those experienced in 2016 may increase if the condition of the AC pipes has deteriorated. As such, we recommend the District undertake a condition assessment of the 150mm AC pipe to identify areas that need pipe replacement.

Project 4 – Watermain Looping (Priority 6-30 years)

There are two options for system looping:

1) Arroyo Road to Grouhel Road, and
2) 12556 Trans Canada Highway to Arroyo Road at Christie Road.

These projects would remove significant dead-ends within the Diamond Improvement District.

Project 5 – 150mm Diameter AC Watermain Replacement Program (Priority 6-30 years)

Based on the findings Projects 1 and 3, the District can implement an AC watermain replacement program to address the aging infrastructure. Due to the large amount of old AC pipe in use, planning should be undertaken to replace the critical portions of the main. All watermain improvements include increasing the number of fire hydrants to meet current standards. There is 2,275m of watermain and it is assumed that all AC pipe will be replaced within 30 years.

7. Capital Expenditure Program

The Local Government Act provides all Improvement Districts with several methods to fund Capital Works. Under Part 17 of the Local Government Act, there are various methods available to finance the costs of works and services required due to new development. A list of potential charges an improvement district can apply is presented below along with a paraphrased description.

- **Section 698 – Powers that must be exercised by bylaw**
  698 (1) (e) An improvement district board may make bylaws for establishing tolls and other charges, including charges for capital expenditures, payable to the improvement district, and the times of their payment.

- **Section 700 – Subdivision servicing requirements**
  700 (1) (a) (b) - The improvement district board may require the provision and construction of: a water distribution system and a fire hydrant system.

- **Section 701 – Requirements for excess or extended services**
  701 (2) - An improvement district may require the owner of land that is to be subdivided to provide portion of water system that will serve land other than the land being subdivided or developed.

Section 702 – Latecomer chargers and cost recovery for excess or extended services

- *702 (3) – If the improvement district pays all or part of the costs of excess or extended services, it may recover costs by a latecomer charge.*

In general, these funding options are to fund two different types of projects, those that serve:

- existing customers; and,
- future development.

Funding methods are described further in the following sections.

7.1. Projects Servicing Existing Customers

Renewal Reserve Fund

Renewal Reserve Funds allow a Local District to put money aside to finance the cost of renewing capital infrastructure. This saving mechanism can reduce or eliminate the amount of borrowing required to construct future works. Renewal Reserve Funds can be formally established with a bylaw passed by the Board of Trustees and registered with the Inspector of Municipalities. Funding sources typically include the following:

- Operating surpluses;
- Budgeted annual contributions;
- Miscellaneous revenue items;
- Sale of assets;

Short Term Borrowing

The Board of Trustees is authorized to pass borrowing bylaws as required to meet approved expenditures. Short term borrowing is typically used to fund unexpected costs and to provide interim financing for capital projects. Temporary borrowing bylaws are registered with the Inspector of Municipalities and are secured by the toll and taxing powers of the Waterworks District.

Long Term Borrowing

The Local Government Act allows a Waterworks District to borrow money by issue and sale of notes, bonds debentures and other securities. In practice this method of borrowing is arranged through the Municipal Finance Authority and is guaranteed by the Province. Long term borrowing bylaws and associated documents are prepared by the Ministry of Community, Sport and Cultural Development.

7.2. Projects Serving Future Development

Capital Expenditure Charges

The ability of a Waterworks District to serve future development may be restricted by the system's capacity to meet that additional demand. Upgrades to the system that are required to meet this increase in demand, but are not tied to a single specific development project are typically funded through a Capital Expenditure Charge (CEC) Fund.

All proposed watermain and fire hydrant upgrades will support future development. The maximum number of lots is assumed to be 259 or 22% greater than the current number of lots at 201. Therefore, watermain upgrades can be apportioned 78% to current users and 22% to future users.

Subdivision Regulation Bylaw

Subdivision applications made to the Ministry of Transportation are referred to the Waterworks District for review and comment. The District's position to support an application is typically based on capacity available in the existing system and the ability to service newly created lots. The cost of servicing new development is typically allocated to the applicant and not existing users.

Waterworks Districts are empowered to pass bylaws that define servicing standards to be met with new construction and the Approving Officer must not approve a subdivision until services meet those standards. Servicing Standards typically apply to development as well as subdivision and are subject to the following limitations:

- Standards can only apply to works that are included in the Letters Patent (waterworks in the case of The Diamond Improvement District Waterworks District).
- The landowner is 100% responsible for the cost of works required to meet standards that are built on his property and on highway adjacent to his property.

Latecomer Charges

In certain cases, the developer is required to oversize works that are constructed on his property or within adjacent road right of way, or he may be required to construct works that extend beyond his property to serve his development. When excess or extended services also benefit adjacent properties, then the developer has a right to request that the Waterworks District pass a Latecomer's Bylaw that allows him to recoup his additional costs.

7.3. 5-Year Capital Plan

Table 7-1 below provides a detailed 5yr Capital Plan breakdown for the initial recommended works. Class C Cost Estimates are provided in Appendix C.

Table 7-1 – 5-Year Capital Plan

Project	2017/18	2018/19	2019/20	2020/21	2021/22
1) Water Network Flow Assessment & Modelling	$15,050				
2) 100mm Diameter Watermain Replacement	$145,200 Grouhel Road	$345,500 Christie Road, Pictou to Arroyo	$442,700 Christie Road, East of Pictou Road	$141,000 Harbour View and Inkster Road	$383,300 Jones, Paton & Lipton Road
3) 150mm AC Condition Assessment	$20,250				
SUBTOTAL	**$180,500**	**$345,500**	**$442,700**	**$141,000**	**$383,300**
TOTAL					**$1,493,000**

7.4. 6 to 30-Year Capital Projects

There is one project recommended for the 6 to 30-year range: Project 4 - Replacement of the 150mm AC watermains. AC pipe is at then end of its expected service life and, for the purposes of this report, it assumed that all AC pipe will be replaced within 30 years. A summary of cost estimate for Project 4: 150mm AC Watermain Replacement is summarized in Table 7-2 below.

Table 7-2 – Cost Estimate for 150mm Diameter AC Watermain Replacement

Project Location	Cost Estimate
Christie Road, South of Grouhel	$274,700
Grouhel Road, Christie to Chandler	$123,600
Chandler Road	$124,800
Grouhel Road, Chandler to Pictou	$397,000
Pictou Road	$216,600
12556/12560 Island Highway	$62,400
Across Island Highway	$226,200
TOTAL	**$1,425,300**

7.5. Proposed Capital Expenditure Charges

Proposed CEC fees assume that 22% of all proposed works benefit future customers. CECs are typically distributed based on a per person basis. The maximum buildout calculations estimate an additional 127 people will occupy 41 new Suburban Residential Homes and 17 Mobile Homes. Assuming an occupancy density of 2.0 persons/household in a Suburban Residential Home, the occupancy for mobile homes is estimated at 1.5 persons/household. The CEC fees are calculated as follows:

- $ 1.49M Cost of Projects in 5-Year Capital Plan
- $ 1.43M Cost of Projects in 6-30 Year Timeline
- $ 2.92M Total Cost Estimate of proposed projects
- $ 642,000 22% of Total Cost Estimate to CECs
- 127 Total new residents
- $ 5054.99 Estimated CEC per person (22% of Total Cost Estimate ÷ Total New Residents)

- **$10,109.98 Estimated CEC per Suburban Residential Home (2 x CEC per person)**
- **$ 7,617.33 Estimated CEC per Mobile Home (1.5 x CEC per person)**

The current Water Services Agreement with the Town of Ladysmith includes the collection of Development Cost Charges (DCC) Equivalent Charge which “shall be an amount equal to the Development Cost Charge established for water under Ladysmith’s Development Cost Charges Bylaw that would apply to the same class of development for the Services if the development were within Ladysmith.” The Ladysmith DCCs must be taken into consideration for the Diamond Improvement District’s CECs.

8. Closure

It has been a pleasure to work with The Diamond Improvement District Water District. We trust this document meets your requirements. If you have any further inquiries, please do not hesitate to contact the undersigned at your earliest convenience.

Yours very truly,

McElhanney Consulting Services Ltd.

Written By,

Darryl Gordon Tunnicliffe

Digitally signed by Darryl Gordon Tunnicliffe
Date: 2017.04.24 14:50:46 -07'00'

D.Tunnicliffe, P.Eng.

Project Engineer

Reviewed By,

R.Irish, P.Eng.

Division Manager, Engineering

Appendix A – Proclamation, Letters Patent & Modifications & Town of Ladysmith Water Use Agreement.

PROCLAMATION

[L.S.] C. WALLACE,
Lieutenant-Governor.

CANADA:

PROVINCE OF BRITISH COLUMBIA

ELIZABETH THE SECOND, by the Grace of God, of Great Britain, Ireland, and the British Dominions beyond the Seas, QUEEN, Defender of the Faith.

To all to whom these presents shall come — GREETING.

R. E. SOMMERS, *Minister of Lands and Forests.*

WHEREAS by section 50 of the "Water Act" it is provided that the Lieutenant-Governor in Council may incorporate a tract of land and the owners thereof into an improvement district:

And whereas the Lieutenant-Governor in Council has, by Order in Council made pursuant to the said Act, been pleased to order that the tract of land hereinafter described and the owners thereof be incorporated into an improvement district under the said Act, and has made further provision to the tenor and effect hereinafter appearing:

NOW KNOW YE that by these presents We do hereby order and proclaim:—

1. The tract of land in the Oyster Land District and the Victoria Land Registration District, more particularly described as follows: Commencing at the north-west corner of District Lot 22G; thence southerly along the westerly boundary of said District Lot 22G and District Lot 23G to the northerly boundary of the Esquimalt and Nanaimo Railway right-of-way; thence in a south-easterly direction along the northerly boundary of the said right-of-way to the north-west corner of District Lot 80; thence southerly along the westerly boundaries of District Lots 80 and 83 to the south-west corner of said District Lot 83; thence easterly along the southerly boundary of District Lot 83 to the westerly boundary of District Lot 70; thence southerly along the westerly boundaries of District Lots 70 and 73 to the south-west corner of said District Lot 73; thence easterly along the southerly boundary of District Lot 73 to the westerly boundary of District Lot 106; thence southerly, easterly, and northerly along the westerly, southerly, and easterly boundaries of the said District Lot 106 to the south-east corner of District Lot 94; thence easterly and northerly along the southerly and easterly boundaries of District Lot 135 to the south-west corner of District Lot 75; thence in a general easterly direction along the southerly boundary of District Lot 75 to Tyee (Rocky) Creek; thence easterly following said Tyee (Rocky) Creek to the high-water mark of Oyster Harbour; thence continuing in a north-westerly direction along said high-water mark of Oyster Harbour to the point of commencement, and all subdivisions thereof, and all owners of land therein are incorporated into an improvement district under the "Water Act" and subject to the provisions thereof and to the conditions hereinafter contained.

2. The said improvement district shall be known as the "Diamond Crossing Fire Protection District."

3. The objects of the said improvement district shall be the provision of fire protection and the acquisition, maintenance, and operation of works, buildings, and equipment for that purpose and all things incidental thereto.

4. There shall be three Trustees of the said improvement district.

5. The persons qualified to vote at the first election of Trustees shall be Canadian citizens who are twenty-one years of age or older and are entitled to be registered as voters under the "Provincial Elections Act" and are owners of land in the tract of land hereinbefore described, and the persons qualified to be candidates at the said election shall be the persons qualified as aforesaid to vote and their wives and husbands.

6. Mr. John Pollock, of Box 28, R.R. 1, Ladysmith, B.C., shall be the Returning Officer for the first election of Trustees. He shall call a general meeting of the persons who are qualified as aforesaid to vote and shall, at least seven days before the date of the said meeting, post at the nearest post-office and at two other conspicuous places within the area hereinbefore described a notice signed by him giving the date, time, and place of holding the said meeting. The Returning Officer shall be chairman of the said meeting, and subject to the provisions of clause 5 he shall have power to determine whether or not any person who desires to vote is qualified to do so. He shall have power to determine the procedure to be followed at the said meeting and the method of taking the votes. Each qualified voter shall be entitled to cast one vote for each of three candidates. The Returning Officer shall declare the result of the election and shall notify the Comptroller of Water Rights of the said result.

7. The candidate elected as Trustee for whom the greatest number of votes is cast at the general meeting called pursuant to clause 6 hereof shall hold office until the annual general meeting of 1956, the candidate elected for whom the second greatest number of votes is cast shall hold office until the annual general meeting of 1955, and the candidate elected for whom the third greatest number of votes is cast shall hold office until the annual general meeting of 1954, but should there be nominated no more than three candidates for the office of Trustee, or should any two or more candidates receive an equal number of votes, then the Returning Officer shall have power to and shall declare which of the candidates are elected and which shall hold office until the next, the second, and the third succeeding annual general meetings respectively.

8. All subsequent elections of Trustees shall be held at the annual general meetings of the improvement district, and it shall be the duty of the

(OVER)

Trustees to call a general meeting to be held between January 1st and May 1st in each year, except 1953, for the following purposes:—

(*a*) To receive from the Trustees a report on condition of the works and a statement of the financial condition of the improvement district:

(*b*) To discuss with the Trustees any matter relating to the works or finances of the improvement district:

(*c*) To fix the remuneration of the Trustees for the ensuing year:

(*d*) To elect a Trustee to succeed the one whose term of office expires coincident with the holding of such annual general meeting, and to elect a Trustee or Trustees to fill any other vacancy or vacancies that has or have occurred or is or are about to occur among the Trustees:

(*e*) To choose the auditor for the ensuing year.

A special general meeting may be called by the Trustees at any time for the purpose of electing a Trustee or Trustees to fill any vacancy or vacancies among the Trustees, or for the purpose of discussing with the owners any matter or matters which in the opinion of the Trustees should be brought up at a general meeting.

At least seven days' notice of every general meeting shall be given by notice posted up in conspicuous places at the nearest post-office and at the building where the meeting is to be held.

The secretary shall enter in a book provided by the Trustees for this purpose minutes of all matters brought before the meeting and the action taken thereon. At any general meeting every person shall be qualified to vote who is a Canadian citizen and is twenty-one years of age or older and is an owner of land in the improvement district, or the authorized agent of any board or corporation that is an owner of such land, or legal representative of any owner of such land who has died, become insolvent, or insane, and is entitled to be registered as a voter under the "Provincial Elections Act." Every person qualified as aforesaid to vote and the wife or husband of any such person shall be qualified to be a candidate for Trustee of the improvement district. In the event of the right of any person to vote at any general meeting being challenged, the chairman shall have authority to determine whether or not such person is entitled to vote, and the chairman may require such person to make and file with him a statutory declaration showing that the declarant is qualified as aforesaid to vote at such general meeting. Forthwith after the holding of a general meeting the Trustees shall file with the Comptroller of Water Rights a true copy of the minutes of such meeting and copies of all auditor's reports and financial statements presented or discussed at the meeting.

9. All words and phrases given special meaning in section 2 of the "Water Act" shall, where used herein, be ascribed the meaning given them in the said section, unless the context otherwise requires.

IN TESTIMONY WHEREOF We have caused these Our Letters to be made Patent and the Great Seal of Our said Province to be hereunto affixed.

WITNESS, His Honour CLARENCE WALLACE, Lieutenant-Governor of Our said Province, in Our City of Victoria, this twenty-first day of April, in the year of our Lord one thousand nine hundred and fifty-three, and in the second year of Our Reign.

By Command.

R. A. PENNINGTON,
Deputy Provincial Secretary.

VICTORIA, B.C.
Printed by DON MCDIARMID, Printer to the Queen's Most Excellent Majesty.
1953

50-453-3277

Province of
British Columbia

Ministry of
Municipal Affairs

Parliament Buildings
Victoria
British Columbia
V8W 3E1

YOUR FILE.....

OUR FILE 0198623

March 10, 1982

Diamond Improvement District
c/o Mrs. P.A. Irvine
Secretary
Box 747
Ladysmith, B.C.
VOR 2EO

Dear Sirs:

Attached please find copies of reprints from the B.C. Gazette in respect to your Improvement District.

Yours truly,

A.O. Ferguson
pp

A.O. Ferguson,
Administrative Officer,
Administrative Services.

Enc.

Diamond Improvement District

LETTERS PATENT

[L.S.] H. P. BELL-IRVING
Lieutenant Governor

CANADA:

PROVINCE OF BRITISH COLUMBIA

ELIZABETH the SECOND, by the Grace of God, of the United Kingdom, Canada and Her Other Realms and Territories, Queen, Head of the Commonwealth, Defender of the Faith

To all to whom these presents shall come—
Greeting

W. N. Vander Zalm
Minister of Municipal Affairs

WHEREAS the Diamond Improvement District is an improvement district incorporated under the *Water Act* by Letters Patent issued on April 21, 1953:

And whereas it is provided by section 825 of the *Municipal Act* that the Lieutenant Governor in Council may at any time amend the Letters Patent of any improvement district in any respect:

And whereas the Board of Trustees of the Diamond Improvement District has requested that the district be empowered to provide the maintenance and operation of a Community Hall as one of its objects:

Now know ye that by these presents We do order and proclaim that on, from and after the date of these supplementary Letters Patent, the provisions of the maintenance and operation of a community hall and all matters incidental thereto be added as an object to the Diamond Improvement District:

And that the Letters Patent dated April 21, 1953, providing for the incorporation of the Diamond Improvement District be amended by striking out section 3 in its entirety and substituting the following therefor:

"3. The objects of the improvement district shall be the provision of fire protection and the acquisition, maintenance and operation of works, buildings and equipment for that purpose; the acquisition, maintenance and operation of works for waterworks purposes; the provision of street lighting; and the maintenance and operation of a community hall; and all things incidental to the foregoing."

And that the Letters Patent of the Diamond Improvement District be deemed to be amended so as to conform to the premises as and from the date of these supplementary Letters Patent.

In testimony whereof, We have caused these Our Letters to be made Patent and the Great Seal of Our said Province to be hereunto affixed.

Witness, the Honourable Henry P. Bell-Irving, Lieutenant Governor of Our said Province of British Columbia, in Our City of Victoria, in Our said Province, this twenty-third day of January in the year of our Lord one thousand nine hundred and eighty-two, and in the thirtieth year of Our Reign.

By Command.

[L.S.] EVAN WOLFE
Provincial Secretary and Minister of Government Services

Reprinted from *The British Columbia Gazette*, February 18, 1982.

Queen's Printer for British Columbia ©
Victoria, 1982

75-282-224

Copy of Minute Approved January 11th, 1962 - Lieutenant-Governor

report:

THAT Diamond Crossing Fire Protection District is an improvement district under the Water Act and was incorporated by Letters Patent issued on April 21st, 1953 (Order in Council No. 958/53).

THAT the Trustees of the District have petitioned to have the Letters Patent of the District amended to change the name of the District, to increase the number of Trustees from three to five and to include waterworks purpose among the objects of the District.

THAT it appears in the public interest to grant the prayer of the petitioners.

AND TO RECOMMEND:

THAT in accord with Section 54 of the Water Act, being Chapter 405 of the Revised Statutes of British Columbia, 1960, the Letters Patent of Diamond Crossing Fire Protection District be amended as follows:

1. By changing the name of the District as set out in Clause 2 thereof to "Diamond Improvement District";

2. By deleting the word "three" from Clause 4 thereof and substituting the word "five" therefore; and

3. By including among the objects of the District the acquisition, maintenance and operation of works for waterworks purpose and all matters incidental thereto.

DATED this 11th day of January A.D. 1962

"W.D. Black"

Acting Minister of Lands and Forests.

APPROVED this 11th day of January A.D. 1962.

"Frank Richter"

Presiding Member of the Executive Council.

Copy of Minute approved 17th December, 1964 - Lieutenant-Governor.

3522.

report:

THAT Diamond Improvement District is an improvement district under the Water Act and was incorporated by Letters Patent issued on April 21st, 1953 (Order in Council No. 958/53).

THAT the Trustees of the said District have petitioned to have the Letters Patent of the District amended to exclude the land hereinafter set out from the tract of land comprising the District.

THAT it appears in the public interest to grant the prayer of the said petitioners.

AND TO RECOMMEND:

THAT in accord with Section 54 of the Water Act, being Chapter 405 of the Revised Statutes of British Columbia, 1960, the Letters Patent of Diamond Improvement District be amended by excluding from the tract of land comprising the District certain land within the Oyster Land District, Victoria Land Registration District, more particularly described as:

Lot 104 except Plan 398 R.W.

DATED this 17th day of Dec. A.D. 1964.

"W. K. Kiernan"

Minister of Lands, Forests, and Water Resources.

APPROVED this 17th day of Dec. A.D. 1964.

"L. R. Peterson"

Presiding Member of the Executive Council.

Diamond

Copy of Minute approved 21st April, 1965 - Lieutenant-Governor.

1192.

report:

THAT Diamond Improvement District is an improvement district under the Water Act and was incorporated by Letters Patent issued on April 21st, 1953 (Order in Council No. 958/53).

THAT the Trustees of the said District have petitioned to have the Letters Patent of the District amended to exclude the land hereinafter set out from the tract of land comprising the District.

THAT it appears in the public interest to grant the prayer of the said petitioners.

AND TO RECOMMEND:

THAT in accord with Section 54 of the Water Act, being Chapter 405 of the Revised Statutes of British Columbia, 1960, the Letters Patent of Diamond Improvement District be amended by excluding from the tract of land comprising the District certain land within the Oyster Land District, Victoria Land Registration District, more particularly described as:

That part of Lot 104 included within the boundaries of Plan 398 R.W.

DATED this 20th day of April A.D. 1965.

"L. R. Peterson"

A/ Minister of Lands, Forests, and Water Resources.

APPROVED this 20th day of April A.D. 1965.

"W. A. C. Bennett"

Presiding Member of the Executive Council.

Diamond Imp Dist

Copy of Minute Approved June 29, 1967 - Lieutenant-Governor

report:

THAT Diamond Improvement District is an improvement district under the Water Act and was incorporated by Letters Patent issued on April 21st, 1953 (Order in Council No. 958/53).

THAT the Trustees of the District have petitioned to have the Letters Patent amended as hereinafter set out.

THAT it appears in the public interest to grant the prayer of the said petitioners.

AND TO RECOMMEND:

THAT in accord with Section 54 of the Water Act, being Chapter 405 of the Revised Statutes of British Columbia, 1960, the Letters Patent of Diamond Improvement District be amended to include among the objects of the District the provision of street lighting and all things incidental thereto.

DATED this 29th day of June A.D. 1967.

"D.L. Brothers"

Acting Minister of Lands, Forests, and Water Resources.

APPROVED this 29th day of June A.D. 1967.

"W.A.C. Bennett"

Presiding Member of the Executive Council.

ADDRESS ALL COMMUNICATIONS TO:
COMPTROLLER OF WATER RIGHTS
PARLIAMENT BUILDINGS, VICTORIA, B.C.

THE GOVERNMENT OF
THE PROVINCE OF BRITISH COLUMBIA

PLEASE REFER TO FILE No. 0198623B

DEPARTMENT OF LANDS, FORESTS, AND WATER RESOURCES
WATER RESOURCES SERVICE
WATER RIGHTS BRANCH
VICTORIA, B.C.

12 June 1973

Diamond Improvement District,
c/o Mrs. P.A. Irvine,
R.R. #1,
#1 - Rocky Creek Rd.,
Ladysmith, B.C.

Dear Sirs:

As requested, please find enclosed copy of District's Letters Patent, 21 April 1953, and copies of amendments dated 11 January 1962, 17 December 1964, 21 April 1965 and 29 June 1967 respectively.

Yours truly,

H.D. DeBeck,
Comptroller of Water Rights

per: K.C. Jolley

KCJ/ii
Encls.

WATER SERVICES AGREEMENT

THIS AGREEMENT is dated for reference the 1st day of July, 2016.

BETWEEN:

TOWN OF LADYSMITH, a municipal corporation incorporated under the *Local Government Act*, PO Box 220, 410 Esplanade, Ladysmith, British Columbia, V9G 1A2

("**Ladysmith**")

AND:

DIAMOND IMPROVEMENT DISTRICT, an improvement district incorporated under the *Local Government Act*, Box 747, Ladysmith, British Columbia, V9G 1A5

("**Diamond**")

GIVEN THAT:

A. Ladysmith is a municipality incorporated under the provisions of the *Local Government Act;*

B. Ladysmith owns and operates the Ladysmith Water System to serve the Town of Ladysmith;

C. Diamond is an Improvement District incorporated under the provisions of the *Local Government Act*;

D. Diamond owns and operates the Diamond Water System to serve the Diamond Improvement District;

E. Ladysmith and Diamond entered into an agreement for the supply of water by Ladysmith to Diamond dated October 2, 1995 that ended October 15, 2015 and since that date, Diamond has been paying Ladysmith the user rates payable under that agreement;

F. This Agreement replaces the agreement referenced in Recital E and settles all outstanding issues with respect to the supply of water by Ladysmith to Diamond;

G. Ladysmith has the authority to enter into an agreement with Diamond to provide water Services to Diamond under section 23 of the *Community Charter;*

H. Ladysmith and Diamond wish to enter into this Agreement to set out the terms and conditions under which Ladysmith will provide the water service to Diamond;

NOW THEREFORE, this Agreement witnesses that in consideration of the premises and of the mutual covenants and Agreements contained in this Agreement, the Parties agree as follows:

1.0 DEFINITIONS AND INTERPRETATION

Definitions

1.1 In this Agreement,

(a) "**Bulk Meter**" means a bulk water service meter to measure the total water supplied as part of the Services, and includes any ancillary vault structure;

(b) "**Business Day**" means any day except Saturdays, Sundays and British Columbia statutory holidays;

(c) "***Community Charter***" means the *Community Charter,* SBC 2003, c. 26;

(d) "**Connection**" means those parts of the Works that connect the Ladysmith Water System to the Diamond Water System

(e) "**DCC Costs Equivalent Charge**" means the charge for water treatment and supply from the Ladysmith Water System required to be paid by Diamond to Ladysmith under Part 5 of this Agreement;

(f) "***Development Cost Charges Bylaw***" means Ladysmith's *Development Cost Charges Bylaw, 2011, No. 1762;*

(g) "**Diamond Water System**" means the Water System owned and operated by Diamond, and, with the exception of the Bulk Meters and Connections, located within the boundaries of Diamond as detailed in the map attached as Schedule A;

(h) "**Director of Infrastructure Services**" means the individual designated by Ladysmith's Council as the Director of Infrastructure Services, or individuals designated by the Council from time to time to act in the place of him or her;

(i) "**Event of Default**" means an event described in Section 14.1;

(j) "**Force Majeure**" means an act of God, act of Canada's enemies, sabotage, war, blockades, insurrections, riots, epidemics, lighting, earthquakes, floods, storms, fires, washouts, nuclear and radiation activity or fall out, arrests and distraints of rulers and people, civil disturbances, explosion, expropriation, or any act, omission or event whether of the kind enumerated in this definition or otherwise not within the control of a Party, which by the exercise of reasonable due diligence, the Party could not have prevented;

(k) "**Ladysmith Water System**" means the Water System owned and operated by

Ladysmith and located within the boundaries of Ladysmith;

(l) "***Local Government Act***" means the *Local Government Act,* RSBC 2015, c. 1;

(m) "**Municipal Master Standards**" means the Master Municipal Construction Document, as amended from time to time, published by the Master Municipal Construction Document Association, subject to amendments made by bylaw or resolution of Ladysmith Council from time to time;

(n) "**Parcel**" means any lot, block or other area of land with its own parcel identifier number within Diamond that can be connected to the Diamond Water System;

(o) "**Parcel Tax Equivalent**" means the amount payable by Diamond to Ladysmith under Part 4 of this Agreement;

(p) "**Party**" means each of Ladysmith and Diamond and its respective successors and permitted assigns, and "**Parties**" means both of them;

(q) "**Reference Date**" means the date indicated on the first page of this Agreement, namely, July 1, 2016;

(r) "**Services**" means the supply of water by Ladysmith to Diamond under this Agreement;

(s) "**Term**" means the period of time set out in Section 8.1;

(t) "***Water Parcel Tax Bylaw***" means Ladysmith's *Water Parcel Tax Bylaw No. 1906;*

(u) "**Water System**" means the water Works and services that are constructed and installed by a Party to distribute potable water and that are owned and operated by that Party; and

(v) "**Works**" means all pipes, valves, pumps, fittings, and other infrastructure comprising the Water System, or required under this Agreement or applicable enactments.

Interpretation

1.2 For the purposes of this Agreement, except as otherwise expressly provided or as the context otherwise requires:

(a) "**Agreement**" means this Agreement, including any schedule, and includes any amendments made to this Agreement by the Parties;

(b) reference in this Agreement to a Part, Section or Schedule is a reference to the corresponding Part, Section or Schedule of this Agreement unless otherwise indicated;

(c) the word "**including**", means including but not limited to;

(d) all references to currency mean Canadian currency;

(e) words in the singular include the plural and words importing a corporate entity include individuals and vice-versa;

(f) reference in this Agreement to an enactment is a reference to an enactment as defined in the *Interpretation Act* (British Columbia), and includes a reference to an enactment of Canada, British Columbia, Ladysmith or Diamond, as applicable; and

(g) reference in this Agreement to an enactment is a reference to that enactment as amended, revised, consolidated or replaced.

Governing Law

1.3 This Agreement shall be governed by and interpreted in accordance with the laws of the Province of British Columbia and the laws of Canada.

Headings

1.4 The headings given to Parts and Sections in this Agreement are for convenience of reference only and do not form part of this Agreement and must not be used in the interpretation of this Agreement.

Severance

1.5 If any clause or portion of this Agreement is declared or held invalid for any reason, the invalidity does not affect the validity of the remainder of that clause or this Agreement, and the terms and provisions of this Agreement continue to be in force and in effect and are to be construed as if the Agreement had been executed without the invalid portion.

2.0 PROVISION OF WATER SERVICES

2.1 Subject to the terms and conditions of this Agreement, Ladysmith agrees to supply bulk water to Diamond up to the following amounts for the corresponding periods:

Year	**Amount**
July 1, 2016 – June 30, 2017	70,000 cubic meters
July 1, 2017 – June 30, 2018	70,000 cubic meters
July 1, 2018 – June 30, 2019	70,000 cubic meters
July 1, 2019 – June 30, 2020	70,000 cubic meters
July 1, 2020 – June 30, 2021	70,000 cubic meters.

2.2 The Parties acknowledge and agree that while Ladysmith is not obligated to supply more water than the amounts detailed in Section 2.1, actual amounts of water supplied by Ladysmith under this Agreement may exceed those amounts and Diamond shall be obligated to pay for any excess water supplied in accordance with Section 3.1.

2.3 Ladysmith shall provide the Service from the Ladysmith Water System to the Diamond Water System at the Connections.

2.4 Ladysmith owns the Ladysmith Water System and shall be solely responsible for its construction, repair, replacement and maintenance, at its sole cost.

2.5 Diamond owns the Diamond Water System and shall be solely responsible for its construction, repair, replacement and maintenance, at its sole cost.

3.0 RATES AND PAYMENT

3.1 The parties agree that Diamond shall pay Ladysmith for the Services at a rate of $0.62 per cubic meter of water supplied to Diamond, for the Term.

3.2 In addition to the amount payable under Section 3.1, Diamond shall pay to Ladysmith:

(a) the Parcel Tax Equivalent in accordance with Part 4;

(b) the DCC Costs Equivalent Charges in accordance with Part 5;

(c) any costs incurred by Ladysmith under Section 10.6; and

(d) interest in respect of arrears of payments owing, in accordance with Section 3.6.

3.3 Diamond shall pay the amount owing under Section 3.1 and Subsection 3.2(a) quarterly. Ladysmith shall invoice Diamond for the amounts owing and Diamond shall pay to Ladysmith the amount invoiced within thirty (30) days of the date of the invoice. The invoice shall set out the details of the calculation used to determine the amount owing for the prior quarter.

3.4 Diamond shall pay the amount owing under Subsection 3.2(b) in accordance with Section 5.4.

3.5 In respect of the payments due and owing under Subsections 3.2 (c) and (d), Ladysmith shall invoice Diamond when amounts become due and owing under this Agreement and Diamond shall pay to Ladysmith the amount invoiced within thirty (30) days of the date of the invoice. The invoice shall set out the details of the calculation used to determine the amount owing.

3.6 Overdue payments owing by Diamond to Ladysmith under this Agreement shall bear interest at a rate of ten percent (10%) per year, calculated annually.

3.7 If a Party has a concern with the amounts paid or invoiced under this Agreement, the Party may request a review of amounts payable, and the Parties shall meet as soon as reasonably possible and attempt to resolve the concerns. The Parties may implement the process in Part 14 if necessary. In connection with such review, the Parties shall each provide such documentation and disclosure as is reasonably necessary to calculate and confirm amounts paid or payable under this Part.

4.0 PARCEL TAX EQUIVALENT

4.1 Without limiting amounts due and owing under other provisions of this Agreement, Diamond shall pay Ladysmith the Parcel Tax Equivalent quarterly as provided in Section 4.2.

4.2 The Parcel Tax Equivalent payable by Diamond to Ladysmith shall be the amount equal to 80% of the water parcel tax payable for a parcel under Ladysmith's *Water Parcel Tax Bylaw* on July 1 of each year of the Term, multiplied by the number of Parcels in Diamond on that date and divided by four to determine the amount to be paid quarterly for the following twelve months.

5.0 DCC COSTS EQUIVALENT

5.1 Without limiting amounts due and owing under other provisions of this Agreement, Diamond shall collect and pay to Ladysmith a DCC Costs Equivalent Charge for water treatment and supply, for each new Parcel or new dwelling connected to the Water System during the Term, as provided in Sections 5.2 to 5.4 below, as a condition of supplying water to the new Parcel or dwelling, as the case may be.

5.2 The DCC Costs Equivalent Charge shall be an amount equal to the Development Cost Charge established for water under Ladysmith's *Development Cost Charges Bylaw* that would apply to the same class of development for the Services if the development were within Ladysmith.

5.3 Diamond shall collect the DCC Costs Equivalent Charge as follows:

(a) at the time of subdivision approval, a DCC Costs Equivalent Charge shall be charged and collected from the subdivision applicant for each Parcel created by the subdivision;

(b) at the time of building permit issuance for the construction of a dwelling, including a secondary suite in an existing dwelling, a DCC Costs Equivalent Charge shall be charged and collected from the owner of the Parcel,

provided that a DCC Costs Equivalent Charge shall not be charged and is not payable where a Development Cost Charge would not be payable under the *Development Cost Charges Bylaw* if the development were in Ladysmith, including where the DCC Costs Equivalent Charge has been previously paid on the same development.

5.4 Diamond shall pay the DCC Costs Equivalent Charge to Ladysmith forthwith when Diamond collects each DCC Costs Equivalent Charge, in each year of the Term of this Agreement. Ladysmith acknowledges and agrees that Diamond is not involved in building permitting, and intends to collect the DCC Costs Equivalent Charge- if it was not collected at subdivision approval- at the time a land owner requests the installation of the water metre required for a dwelling, which is typically close to the time of building permit issuance.

5.5 Without limiting Section 8.6, the Parties shall undertake an annual reconciliation and adjustment of the DCC Costs Equivalent Charges payable under this Part 5 in each year of the Term of this Agreement, which shall be completed by February 28 each year for the preceding year. In connection with such reconciliation and adjustment, the Parties shall each provide such documentation and disclosure as is reasonably necessary to calculate and confirm the DCC Costs Equivalent Charges and in connection with this documentation and disclosure, Diamond shall provide a detailed report to Ladysmith of all subdivisions approved and construction of dwellings within Diamond in the preceding year. Any disagreements or disputes respecting the determination of amounts payable under this Part 5 shall be resolved in accordance with the procedures set out in Part 14.

6.0 WATER QUALITY, QUANTITY AND SERVICE INTERRUPTIONS

6.1 The water supplied to Diamond by Ladysmith shall be of substantially the same quality as water provided to other users of the Ladysmith Water System. Ladysmith shall immediately notify Diamond of any water quality issues or concerns that arise during the Term, provide to Diamond a copy of any water quality testing results related to that concern, take all steps reasonably necessary to resolve any water quality concerns as soon as possible after such a concern is identified, and shall keep Diamond informed of the steps being taken and the anticipated timeline for resolution.

6.2 Ladysmith shall provide the Services at the same level or degree to which water services are provided elsewhere within Ladysmith. Ladysmith makes no representation or warranty that the level or degree of the Services provided under this Agreement will be maintained or continued to any particular standard, other than as stated expressly herein. For greater certainty, in the event that the Ladysmith is subject to water restrictions, Diamond agrees that the same water restrictions will also apply to Diamond and Diamond shall take all reasonable and necessary actions to effect those water restrictions within the Diamond Improvement District.

6.3 Ladysmith represents and warrants that the water provided under this Agreement meets British Columbia drinking water guidelines at the Connections, and Diamond agrees that it is responsible for ensuring that water distributed within the Diamond Water System shall also meet British Columbia drinking water guidelines through the point of delivery to its customers.

6.4 Diamond acknowledges and agrees that there may be from time to time interruptions or reductions in the level of the Services, and that Ladysmith shall not be held liable for any

losses, costs, damages, claims or expenses arising from or connected with a temporary interruption or reduction in the level of the Services provided under this Agreement.

6.5 Ladysmith shall provide Diamond with at least fifteen (15) days advance notice of all planned interruptions to the Services, and shall advise Diamond immediately of all unplanned interruptions. Ladysmith shall use best efforts to minimize all planned interruptions, and to ensure that planned and unplanned service interruptions are addressed as quickly as possible in order to restore the Services as expeditiously as reasonably possible. Ladysmith shall provide periodic updates to Diamond during all service interruptions respecting the anticipated timeframe for restoring the Services.

7.0 REPRESENTATIONS AND WARRANTIES

7.1 Each Party represents and warrants to the other as follows:

(a) it has the power and capacity to enter into this Agreement;

(b) this Agreement is valid and binding on it in accordance with its terms;

(c) the performance of its obligations under this Agreement does not breach the terms of any other agreement or obligation to which it is a party; and

(d) it now holds and will hold its Water System beneficially and free and clear of all restrictions, liens, encumbrances, floating charges or financial obligations of any kind except:

(i) the obligations created under this Agreement; or

(ii) as agreed between the Parties from time to time.

8.0 TERM AND TERMINATION

8.1 This Agreement takes effect on the Reference Date and is for a term of five (5) years, ending on June 30, 2021, unless terminated earlier in accordance with this Agreement.

8.2 At least one (1) year prior to the end of the Term, the Parties shall negotiate and make best efforts to reach agreement on an extension or replacement of this Agreement.

8.3 The Parties may terminate this Agreement by mutual agreement.

8.4 A Party may terminate this Agreement by providing at least six (6) month's written notice to the other Party if there has been an Event of Default that has not been resolved through the dispute resolution process set out in Part 14, or the Parties are unable to amend this Agreement in accordance with Section 16.2.

8.5 Despite any other provisions of this Agreement, neither Party may terminate this Agreement during any attempt to resolve issues through the dispute resolution process set out in Part 14.

8.6 If this Agreement is terminated or comes to an end, there shall be a reconciliation and final adjustment of amounts owing by Diamond to Ladysmith under Part 3, calculated up to the date of termination, and the obligation to make any adjusting payment shall survive the termination of this Agreement.

8.7 If Ladysmith continues to provide the Services to Diamond following the Term without any further written agreement, the terms and conditions set out in this Agreement, including all amounts payable under Part 3, shall continue to apply on a month to month basis until a new agreement is made or the Services are terminated, whichever occurs first, provided that the user rate payable under section 3.1 shall be equivalent to 90% of the rate then applicable for residential users in Ladysmith.

9.0 AREA AND CONNECTION TO THE LADYSMITH WATER SYSTEM

9.1 There is no restriction under this Agreement as to the areas within Diamond that may be connected to the Diamond Water System, provided that Diamond agrees that the Services shall not be resold, connected or provided to any person or entity, directly or indirectly, other than to a Parcel, person or entity within the boundaries of Diamond.

9.2 During the Term, Diamond agrees to notify Ladysmith of any change in the boundaries of Diamond.

9.3 Diamond agrees not to supply or provide the Diamond Water System with any source of water other than the water provided by Ladysmith under this Agreement, unless otherwise agreed by the Parties in writing.

9.4 The Parties acknowledge that the supply of water by Ladysmith to Diamond is currently supplied by two Connections between the Ladysmith Water System and the Diamond Water System as shown in Schedule A.

9.5 The Parties acknowledge and agree that the locations of Connections existing on the Reference Date are acceptable to both parties. Ladysmith agrees that it shall not require the relocation of existing Connections without first consulting with Diamond and if it requires relocation Diamond shall be responsible for all risk and costs associated with such relocation.

9.6 The maintenance, repair and replacement, if necessary, of the Connections shall be at the sole expense of Diamond.

9.7 Subject to Section 9.8, Diamond may construct, install and use one or more additional Connections if:

(a) Diamond pays for all costs related to designing, financing, constructing, installing, maintaining and repairing the additional Connection;

(b) Diamond pays for all costs related to any upgrades to Ladysmith's transmission mains necessitated by the additional Connections; and

(c) before commencing construction of an additional Connection, Diamond submits all plans, specifications and design flows to Ladysmith, as requested by the Director of Infrastructure Services, to ensure the additional Connection conforms to Municipal Master Standards and the requirements for any Bulk Meter to be installed as part of the additional Connection, and Ladysmith approves in writing such plans, specifications and design flows.

9.8 The number of additional Connections permitted during the Term shall be limited to a maximum of 10 Connections, unless otherwise agreed in writing by the Parties. Diamond agrees that any additional Connections shall not obligate Ladysmith to provide water in greater quantities than set out in Section 2.1.

9.9 The parties acknowledge that Ladysmith currently supplies water to the Stz'uminus First Nation under a separate water services agreement, and Diamond agrees that it shall not connect to or obtain water from the water main that provides water from Ladysmith to the Stz'uminus First Nation without the written approval of Ladysmith.

10.0 METERING

10.1 The Parties acknowledge and agree that the locations of the two (2) Bulk Meters existing on the Reference Date are acceptable to Ladysmith and the Director of Infrastructure Services. Ladysmith agrees that it shall not require the relocation of the existing Bulk Meters without first consulting with Diamond and shall not require the replacement of a Bulk Meter that is operational without first consulting with Diamond.

10.2 The Parties agree that Diamond is responsible for the installation, operation, maintenance, repair, and replacement if necessary, of the Bulk Meters. Ladysmith hereby grants to Diamond a non-exclusive license to access and occupy the locations of the Bulk Meters at all times during the Term, for these purposes.

10.3 Ladysmith shall measure the water flow from the Ladysmith Water System to the Diamond Water System by reading each Bulk Meter and adding together the amount from each Bulk Meter to determine the total amount of water used by Diamond.

10.4 Either Ladysmith or Diamond at its sole cost may at any time appoint a qualified representative to ascertain the accuracy of any Bulk Meter used in connection with this Agreement.

10.5 Ladysmith shall notify Diamond of any repairs or maintenance that it determines are required to the Bulk Meter or the Connections, and on being notified Diamond shall take

all reasonable steps to complete the work and shall advise Ladysmith of the progress and completion of the work.

10.6. If at any time Ladysmith determines that emergency repairs are required to a Bulk Meter or Connection, Ladysmith shall contact the Diamond Emergency Contacts by telephone to discuss the emergency, the nature of the repairs required, and an appropriate timeline for completing those repairs. If Ladysmith is unable to reach either of the Diamond Emergency Contacts within 12 hours, or if Diamond advises that it is unable to make the necessary immediate repairs required, Ladysmith may undertake such repairs at the cost of Diamond. Ladysmith shall undertake only such repairs as necessary to address the emergency situation. The Diamond Emergency Contacts are: Mr. Jeremy Kaye, Trustee, (250) 714-3822 (c) and (250) 245-3844 (h); and Ms. Debby Baker, Administrator, (250)740-5259 (c) or such other person(s) as Diamond may advise Ladysmith of in writing.

11.0 ON-GOING COMMUNICATIONS

11.1 Each Party shall, within ten (10) days of the commencement of this Agreement, appoint a representative as the primary contact person for all matters arising under this Agreement and provide contact information for that person to the other party.

11.2 The representatives shall

(a) oversee the implementation of this Agreement and the delivery of the Services;

(b) meet as needed to review and discuss any issues arising under this Agreement;

(c) each be available and respond in a timely manner to an inquiry from the other representative respecting this Agreement;

(d) review and attempt to resolve any Service delivery issues that arise from time to time, including Service interruptions;

(e) consider and recommend amendments to this Agreement to address changes in circumstances or other issues that arise; and

(f) each bring forward issues and concerns as necessary with their respective Council and Board of Trustees for input and direction.

12.0 INDEMNITY AND INSURANCE

12.1 Diamond releases, indemnifies and saves harmless Ladysmith from and against all claims, demands, actions, suits, loss, damage, costs (including legal costs), fines, penalties, charges and expenses (in this Part, collectively "**Claims**") which Ladysmith may incur, suffer or be put to arising out of or in connection with this Agreement or the provision of the Services, directly or indirectly, with respect to the following:

(a) Claims arising out of or in connection with any loss or damage to persons (including bodily injury or death) or property within the boundaries of Diamond; and

(b) Claims arising from a breach by Diamond of any of its agreements, representations, warranties or covenants set forth in this Agreement.

12.2 Ladysmith releases, indemnifies and saves harmless Diamond from and against all Claims, which Diamond may incur, suffer or be put to arising out of or in connection with this Agreement or the provision of the Services, directly or indirectly, with respect to the following:

(a) Claims arising out of or in connection with any loss or damage to persons (including bodily injury or death) or property within the boundaries of Ladysmith; and

(b) Claims arising from a breach by Ladysmith of any of its agreements, representations, warranties or covenants set forth in this Agreement.

12.3 The indemnities in Sections 12.1 and 12.2 are:

(a) subject to Section 14.5; and

(b) survive the expiration or termination of this Agreement.

12.4 Each Party shall take out and keep in force during the Term comprehensive general liability (including bodily injury, death, property damage, and damage or loss) insurance on an occurrence basis with respect to its Water System and operations of not less than $5,000,000.00 per occurrence (or such greater reasonable amount as the Parties may agree to from time to time), which insurance shall include the other Party as a named insured and shall protect the other Party in respect of claims as if that Party were separately insured.

12.5 The insurance required to be maintained by each Party shall be on terms and with insurers to which the other Party has no reasonable objection and shall require the insurer to provide to the other Party thirty (30) days' prior written notice of cancellation or material alteration of its terms.

12.6 At the time of execution of this Agreement and at other reasonable times requested by a Party, the other Party shall furnish evidence of insurance to the requesting Party, either by means of a certified copy of the policy or policies of insurance with all amendments and endorsements or a certificate from that Party's insurer which, in the case of comprehensive general liability insurance, shall provide such information as the other Party reasonably requires.

12.7 If a Party fails to take out, renew, and keep in force the required insurance, then the other Party may do so as the agent of that Party and the costs of doing so shall be a debt due and owing to the Party that obtained such insurance, payable immediately on demand.

13.0 RECORDS

13.1 Each Party shall cause all books and records with respect to this Agreement and its Water System to be kept in its offices. Each Party shall permit the other Party, its accountants and other representatives, at the other Party's own expense and at all reasonable times to examine and make copies of any and all documents under the control of a Party and relating to this Agreement.

14.0 DEFAULT AND DISPUTE RESOLUTION

14.1 The occurrence of any of the following events shall be an Event of Default under this Agreement:

(a) the failure of a Party to perform or observe any of its covenants or agreements in this Agreement, if such failure is not cured within 45 days of written notice from the other Party specifying such failure, or if full compliance is not possible for reasons beyond a Party's control, a failure to provide the other Party with a plan and schedule of compliance and to proceed diligently with the plan in accordance with the schedule;

(b) a Party:

(i) becomes insolvent; or

(ii) acknowledges its insolvency;

(c) a statutory trustee or provincial agent is appointed in respect of any property or asset of a Party and is not discharged within 45 days; or

(d) Diamond fails to pay Ladysmith any of the amounts payable under this Agreement, within three (3) months of written notice to Diamond of failure to pay.

14.2 If the Parties are unable to agree on the interpretation or application of any provision in the Agreement, if there is an Event of Default, or if the Parties are unable to resolve any other issue relating to this Agreement, the Parties agree to the following process in the order it is set out:

(a) the Party initiating the process shall send written notice to the other Party;

(b) the Parties shall promptly, diligently and in good faith take all reasonable measures to negotiate an acceptable resolution to the disagreement or dispute;

(c) if the Parties are unable to negotiate a resolution under paragraph (b) within thirty (30) Business Days, the parties may, by agreement, appoint a mutually acceptable person to mediate the matter, and the parties must act reasonably and in good faith and cooperate with the mediator and with each other in an attempt to resolve the matter within thirty (30) Business Days after the mediator is appointed. The costs of

the mediator and any costs of facilities shall be shared equally by the parties; and

(d) if the Parties do not implement mediation under paragraph (c) or are unable to resolve the dispute if mediation under paragraph (c) is implemented, either Party may give notice to the other Party that the matter shall be decided by binding arbitration.

14.3 Where notice is given under Section 14.2(d), a single arbitrator shall be appointed by agreement within sixty (60) Business Days of the notice, and if the Parties are unable to agree within that time, the arbitrator shall be appointed by the British Columbia International Commercial Arbitration Centre. The arbitration shall follow the rules of the *Arbitration Act,* RSBC 1996 c. 55 unless the Parties agree otherwise, and the decision of the arbitrator is final, binding and conclusive.

14.4 Unless otherwise agreed by the Parties or ordered by an arbitrator, each Party shall pay its own legal and other costs, and shall share equally the costs for an arbitrator and any required facilities.

14.5 No Party shall be liable for its failure to perform any of its obligations under this Agreement due to Force Majeure.

15.0 APPROVALS

15.1 This Agreement is subject to the following approvals and shall come into effect only after both of these approvals are obtained:

(a) the municipal Council of Ladysmith shall have approved this Agreement and satisfied all of its statutory obligations under the *Community Charter, Local Government Act* and other applicable enactment; and

(b) the Board of Trustees of Diamond shall have approved this Agreement and satisfied all of its statutory obligations under the *Local Government Act* and other applicable enactments.

16.0 AMENDMENTS

16.1 This Agreement may not be modified or amended except by written agreement duly authorized by the Parties.

16.2 If any laws of Canada or British Columbia are promulgated, amended or repealed, such that substantial portions of the Agreement are no longer effective or incapable of being performed in the opinion of Diamond or Ladysmith, acting reasonably, the Parties agree to use their reasonable best efforts to negotiate such amendments to this Agreement as are necessary to give effect to the intention of the Parties under this Agreement. Should negotiations fail to result in an agreement being reached within one year of the date of the first meeting, either Party shall have the option of terminating this Agreement in accordance with Section 8.4.

17.0 NOTICES

17.1 Any notice or other communication under this Agreement shall be in writing and shall be given by the delivery or rendering thereof to its addressee by hand, by prepaid first class mail, by email or by facsimile transmission, to the contact information below:

(a) If to Ladysmith:

PO Box 220, 410 Esplanade
Ladysmith, British Columbia, V9G 1A2
Fax: 250-245-6411
Email: gferrero@ladysmith.ca

Attention: Chief Administrative Officer

(b) If to Diamond:

Box 747
Ladysmith, British Columbia, V9G 1A5
Phone: 250-740-5259
Fax: 250-245-9558
Email: admin@didbc.com

Attention: Board Chair

17.2 A notice or other communication so given shall be deemed to have been received at the time of its delivery if delivered by hand, three (3) Business Days after the date of mailing if mailed, if by email or fax, on the next Business Day after it is transmitted. Each Party shall notify the other Party of any change of contact information.

18.0 MISCELLANEOUS

Entire Agreement

18.1 This Agreement is the entire agreement between the parties respecting its subject matter and there are no undertakings, representations or promises, express or implied, other than those expressly set out in this Agreement.

Agreement for Services

18.2 This Agreement is an agreement for the provision of services, and nothing in this Agreement is intended to be or shall be interpreted as creating a partnership, agency or joint venture relationship of any kind between the Parties, or as imposing on either Party a partnership duty, liability or obligation of any kind in relation to the other Party.

Enurement

18.3 This Agreement shall be binding upon and enure to the benefit of the Parties and their respective successors and permitted assigns.

Further Assurances

18.4 Each Party shall perform any act and execute and deliver any document reasonably required by any other Party to carry out the terms of this Agreement in accordance with its true intent and meaning.

Statutory Obligations and Powers Preserved

18.5 The obligations of the Parties under this Agreement are always subject to the *Community Charter*, *Local Government Act*, and other applicable enactments and the Parties shall abide by all applicable enactments in carrying out the terms of this Agreement. Except as specifically provided, nothing contained or implied in this Agreement will impair or affect the Parties' rights and powers in the exercise of their functions pursuant to the *Community Charter*, the *Local Government Act* or any other enactment.

No Assignment

18.6 Neither this Agreement nor any part of it may be assigned by either Party without the written consent of the other Party.

Consents and Waivers

18.7 No consent or waiver, express or implied, by either Party of any breach or default by the other under this Agreement shall:

(a) be valid unless it is in writing and stated to be a consent or waiver pursuant to this Section;

(b) be relied on as a consent to or waiver of any other breach or default of the same or any other nature;

(c) constitute a general waiver under this Agreement; or

(d) eliminate or modify the need for a specific consent or waiver under this Section in any other or subsequent instance.

Time of the Essence

18.8 Time is of the essence in the performance of each obligation under this Agreement.

Survival on Termination

18.9 The provisions of this Agreement necessary to give full effect thereto shall survive the

termination of this Agreement and, despite termination of this Agreement, no Party shall by reason of such termination be relieved of any obligation or liability toward the other Party accrued prior to termination.

Further Agreements

18.10 Subject to any approval of the municipal Council of Ladysmith and Diamond's Board of Trustees, required by statute, each of the Parties shall cooperate fully and take all reasonable steps to negotiate, finalize and execute all agreements, instruments and other documents contemplated by or related to this Agreement.

IN WITNESS WHEREOF the Parties have executed this Agreement as of the date written below.

EXECUTED THIS _____ DAY OF _____, 2016 BY THE **TOWN OF LADYSMITH**, by its authorized signatories:

Per: _____
Mayor

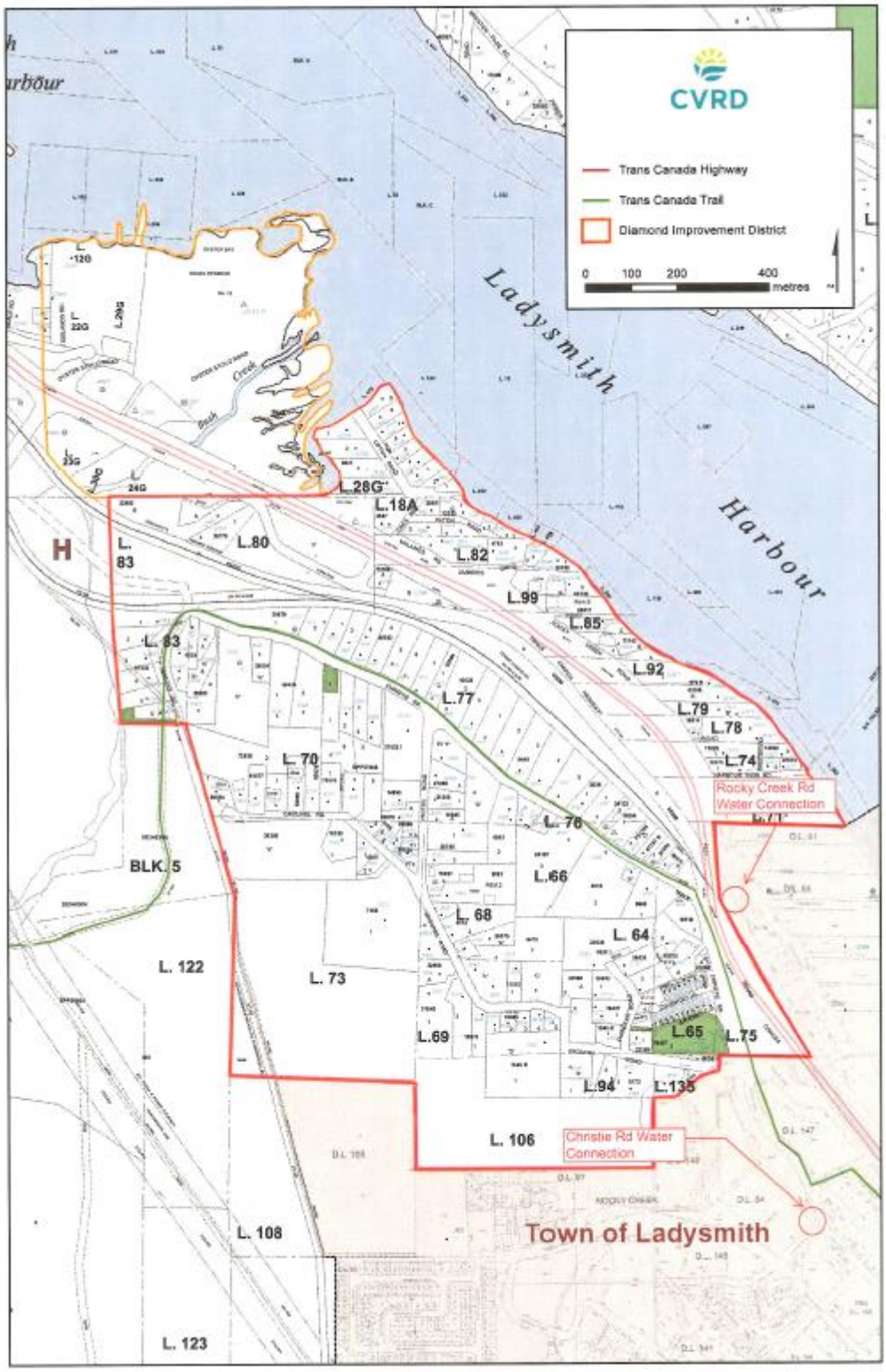
Per: _____
Chief Administrative Officer

EXCECUTED THIS _____ DAY OF _____, 2016 BY THE **DIAMOND IMPROVEMENT DISTRICT**, by its authorized signatories:

Per: _____
Board Chair

Per: _____
Trustee

SCHEDULE A
MAP OF DIAMOND IMPROVEMENT DISTRICT



Appendix B – Hydrant Inspection Reports

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

2013 ANNUAL INSPECTION REPORT

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company Diamond Water Board Date June 14 2013

Hydrant # 14 Fire Dept. # _____ Location 12411 Rocky Creek Rd

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-PSR

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 90 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	(☑)	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks REPLACED THE MAIN VALVE.

Inspector [signature]

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 12 2013

Hydrant # 15 Fire Dept. # _____ Location 4931 HARBOUR VIEW RD.

Hydrant Type: ☐ Mueller _____ ☑ Terminal City TC-2

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size N/A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 96 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks HOSE CAP DAMAGE TO THE MAIN VALVE

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax (250) 754-4299

HYDRANT INSPECTION REPORT

Company Diamond Water Board Date June 14 2013

Hydrant # 16 Fire Dept. # ___ Location 12435 Rocky Creek Rd

Hydrant Type: ☐ Mueller ☑ Terminal City C-71-PSR

☐ Canada Valve ☐ Other

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 105 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	(☑)	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks Replaced main valve.

Inspector

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 14 2013

Hydrant # 17 Fire Dept. # ____ Location 12457 ROCKY CREEK RD

Hydrant Type: ☐ Mueller ____ ☑ Terminal City C-71-BR

☐ Canada Valve ____ ☐ Other ____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 114 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	(☑)	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks REPLACED THE MAIN VALVE

Inspector [signature]

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 14 2013

Hydrant # 18 Fire Dept. # _____ Location 12483 Rocky Creek Rd.

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-PSR

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 116 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks ALL OK.

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 14 2013

Hydrant # 19 Fire Dept. # _____ Location 12505 Rocky Creek Rd.

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-BR

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 114 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☑	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks REPLACED THE MAIN VALVE

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER LTD Date JUNE 11 2013

Hydrant # 20 Fire Dept. # _____ Location TCH & Rocky Creek Rd. S/B SIDE

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-P

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 112 psi Barrel Depth 6 Barrel Drain Time 15 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks VERY HIGH WATER TABLE. THE HYDRANT NEEDED TO BE PUMPED OUT TO REDUCE THE WATER TABLE

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date SEPT. 09 2013

Hydrant # 21 Fire Dept. # ____ Location ROCKY CREEK RD & MALAMUTE CR.

Hydrant Type: ☐ Mueller ____ ☑ Terminal City C-71-PSR

☐ Canada Valve ____ ☐ Other ____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 110 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☐ A ☑ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☐	☐	☑
Main valve	☐	☐	☑
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks NOT ABLE TO SHUT OFF THE WATER TO LOWER DIAMOND TO INSPECT MAIN VALVE

Inspector ____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
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HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 14 2013

Hydrant # 22 Fire Dept. # _____ Location MALAMOS RD.

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-BR

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 114 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	(☑)	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks REPLACED THE MAIN VALVE

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
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HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date OCTOBER 05 2013

Hydrant # 23 Fire Dept. # ___ Location 5071 WILLIAMS RD.

Hydrant Type: ☐ Mueller ☑ Terminal City CITY-BSR.

☐ Canada Valve ☐ Other

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 118 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks

Inspector

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
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HYDRANT INSPECTION REPORT

Company Diamond Water Parks Date June 11 2013

Hydrant # 28 Fire Dept. # _____ Location 12446 Arroyo Rd.

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-PSR

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 92 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks _____

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
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HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 11 2013

Hydrant # 24 Fire Dept. # ____ Location 12556 TRANSCANADA Hwy

Hydrant Type: ☐ Mueller ____ ☑ Terminal City C.F.I.P

☐ Canada Valve ____ ☐ Other ____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☑ Right ☐ Left No. of Turns to Operate 12

Pressure 122 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks MINOR ROCK DAMAGE TO MAIN VALVE AND VALVE SEAT RING.

Inspector [signature]

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
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HYDRANT INSPECTION REPORT

Company DIAMOND LAKE BOATS Date OCTOBER 05 2013

Hydrant # 25 Fire Dept. # _____ Location T.C. HWY NORTH BOUND

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-ASE

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 124 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks _____

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
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HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 12 2013

Hydrant # 26 Fire Dept. # _____ Location 5041 PATON RD

Hydrant Type: ☐ Mueller _____ ☑ Terminal City TC # 2

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4½A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 110 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☑	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☑	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☑	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks REPLACED A CAP GASKET, DRAIN VALVE AND HEAD GASKET.

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
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HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 12 2013

Hydrant # 27 Fire Dept. # _____ Location 12575 LIPTON RD

Hydrant Type: ☐ Mueller _____ ☑ Terminal City TC-2

☐ Canada Valve _____ ☐ Other _____ _____

Hose Port Size 2½" Pumper Port Size N/A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 110 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
~~Hydrant checked for leaks and clearly accessible from the road~~	☑	☐	☐

Remarks _____

_____

_____

Inspector _____

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

2014 ANNUAL INSPECTION REPORT

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 19 2014.

Hydrant # 1 Fire Dept. Hydrant # _____ Location 4995 GROWER PL

Hydrant Type: ☐ Mueller _____ ☑ Terminal City #2

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2.5" Pumper Port Size N/A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 74 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks MINOR ROCK DAMAGE TO MAIN VALVE & DRAIN VALVE ASSEMBLY

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 19 2014

Hydrant # 2 Fire Dept. Hydrant # ____ Location 5053 GRAVEL RD

Hydrant Type: ☐ Mueller ____ ☑ Terminal City # 2

☐ Canada Valve ____ ☐ Other ____

Hose Port Size 2.5" Pumper Port Size N/A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 60 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks ALL OK

Inspector

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
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HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 19, 2014

Hydrant # ~~3~~ 4 Fire Dept. Hydrant # _____ Location ~~5075~~ GRAINGER RD

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-P

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2.5" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 61 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks ALL OK

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company Diamond Water Homes Date June 19 2024

Hydrant # #3 Fire Dept. Hydrant # _____ Location Pictou & Grenfell Pl.

Hydrant Type: ☐ Mueller _____ ☑ Terminal City #2

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2.5" Pumper Port Size N/A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 74 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks All ok.

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ♦ Nanaimo, BC ♦ V9R 4R8
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HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JULY 27 2014

Hydrant # 5 Fire Dept. Hydrant # _____ Location 5060 PICTOU

Hydrant Type: ☐ Mueller _____ ☑ Terminal City #2

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2.5" Pumper Port Size N/A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 80 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks _____

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ♦ Nanaimo, BC ♦ V9R 4R8
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HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date June 27 2014.

Hydrant # 6 Fire Dept. Hydrant # ____ Location 5100 CHRISTIE

Hydrant Type: ☐ Mueller ____ ☑ Terminal City TC #2

☐ Canada Valve ____ ☐ Other ____

Hose Port Size 2.5" Pumper Port Size 4½

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 96 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks ____

Inspector ____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date OCT. 25. 2014

Hydrant # 7 Fire Dept. Hydrant # _____ Location ENTRANCE TO CLOVER MHP

Hydrant Type: ☐ Mueller _____ ☑ Terminal City #2

☐ Canada Valve _____ ☐ Other _____

Hose Port Size _____ Pumper Port Size W/O

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure N/A psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	(☑)	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks REPLACED MAIN VALVE

Inspector [signature]

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 27 2014

Hydrant # 8 Fire Dept. Hydrant # _____ Location 5020 CHRISTIE

Hydrant Type: ☐ Mueller _____ ☑ Terminal City #2

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2.5" Pumper Port Size N/A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 88 psi Barrel Depth _____ Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks MAJOR ROCK DAMAGE TO MAIN VALVE.

Inspector [signature]

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date DEC 27 2014.

Hydrant # 9 Fire Dept. Hydrant # _____ Location 4990 CHRISTIE RD.

Hydrant Type: ☐ Mueller _____ ☑ Terminal City # 2

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2.5" Pumper Port Size N/A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 78 psi Barrel Depth _____ Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	(☑)	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main ~~stem/safety stem~~ coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
~~Drain valve~~	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks Replaced CAP GASKET

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARDS Date JUNE 19 2014

Hydrant # 10 Fire Dept. Hydrant # _____ Location WATER BOARDS CHRISTIE RD. HALL

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-P

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2.5" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 76 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks _____

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date July 18 2014

Hydrant # 11 Fire Dept. Hydrant # _____ Location ~~12300~~ 4926 CHRISTIE RD

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-PSE

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2.5" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 82 psi Barrel Depth 10' Barrel Drain Time 15 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☐ Yes ☐ No

Service Procedure List

Type of Service: ☐ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks CLEARED BRUSH FROM AROUND THE HYDRANT.

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 19. 2014

Hydrant # 12 Fire Dept. Hydrant # ____ Location COURTNEY & CHRISTIE RDS.

Hydrant Type: ☐ Mueller ____ ☑ Terminal City C-71-PSR

☐ Canada Valve ____ ☐ Other ____

Hose Port Size 2.5" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 84 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks ____

Inspector [signature]

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road ◆ Nanaimo, BC ◆ V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date June 27 2014

Hydrant # 13 Fire Dept. Hydrant # _____ Location 12281 CHANDLER RD.

Hydrant Type: ☐ Mueller _____ ☑ Terminal City TC #2

☐ Canada Valve _____ ☐ Other _____

Hose Port Size _____ Pumper Port Size N/A

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 27

Pressure 80 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks _____

Inspector _____

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company Diamond Water Board Date Nov. 15. 2014

Hydrant # 2425 Fire Dept. Hydrant # _____ Location 12556 T.C. Highway

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-P

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☑ Right ☐ Left No. of Turns to Operate 12

Pressure 120 psi Barrel Depth 6' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	(☑)	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks Replaced main valve. Requested to inspect hydrant after flushing by water board.

Inspector [signature]

Guaranteed Quality and Service

PACIFIC FIRE HYDRANT INSPECTION SERVICE

860 Victoria Road • Nanaimo, BC • V9R 4R8
Phone and Fax: (250) 754-4299

HYDRANT INSPECTION REPORT

Company DIAMOND WATER BOARD Date JUNE 19 2014

Hydrant # 28 Fire Dept. Hydrant # _____ Location 12445 ARROYO RD.

Hydrant Type: ☐ Mueller _____ ☑ Terminal City C-71-PSR.

☐ Canada Valve _____ ☐ Other _____

Hose Port Size 2½" Pumper Port Size 4"

Direction of Operation: ☐ Right ☑ Left No. of Turns to Operate 12

Pressure 92 psi Barrel Depth 8' Barrel Drain Time 10 mins

Hydrant Flushed: ☑ Yes ☐ No Did Water Clear: ☑ Yes ☐ No

Service Procedure List

Type of Service: ☑ A ☐ B

	Passed	Failed (repaired)	N/A
Fully opened, checked for leaks at ports, top end and drainage	☑	☐	☐
Bonnet and bolts	☑	☐	☐
Port caps and gaskets	☑	☐	☐
Port and pumper port threads	☑	☐	☐
Packing glands/top end oil	☑	☐	☐
Top flange assembly/hold down nut	☑	☐	☐
Main stem/safety stem coupling	☑	☐	☐
Main valve assembly	☑	☐	☐
Main valve	☑	☐	☐
Drain valve	☑	☐	☐
Hydrant checked for leaks and clearly accessible from the road	☑	☐	☐

Remarks ALL OK

Inspector [signature]

Guaranteed Quality and Service

Appendix C – Class C Cost Estimates

McElhanney Consulting Services Ltd.
2231-31902-01

DETAILED COST ESTIMATES

February 7, 2017

Project 1 – Water Network Flow Assessment & Modelling (Priority 1-5 years)

Description	QTY	Units	Rate	Extension	Total
Hydrant Flow Testing & Analysis	8	hours	$ 180.00	$ 1,440.00	
Computational Modeling & Report	1	LS	$ 8,250.00	$ 8,250.00	
					$ 9,690.00

Project 2 – 150mm AC Condition Assessment (Priority 1-5 years)

Description	QTY	Units	Rate	Extension	Total
Isolated Pressure Testing	16	hours	$ 180.00	$ 2,880.00	
Summary Report	1	LS	$ 750.00	$ 750.00	
					$ 3,630.00

Project 3 – 100mm Diameter Watermain Replacement (Priority 1-5 years)

Description	QTY	Units	Rate	Extension	Total
150 mm dia. PVC Pipe	2430	m	$ 275.00	$ 668,250.00	
150mm gate valves	24	ea	$ 1,500.00	$ 36,000.00	
hydrants	15	ea	$ 5,500.00	$ 82,500.00	
50mm asphalt cement, 1.2m wide	245	m	$ 42.00	$ 10,290.00	
75mm base gravel, 1.2m wide	245	m	$ 30.00	$ 7,350.00	
150mm subbase gravel, 1.2m wide	245	m	$ 72.00	$ 17,640.00	
shoulder gravel, 1.0m wide	2430	m	$ 25.00	$ 60,750.00	
standpipe	2	ea	$ 350.00	$ 700.00	
tie-in to existing watermain	10	ea	$ 7,500.00	$ 75,000.00	
				Subtotal	$ 958,480.00
				Contingency (30%)	$ 287,544.00
				Subtotal	$ 1,246,024.00
				Engineering (15%)	$ 186,903.60
				Total	**$ 1,432,927.60**
				Total Price/meter	**$ 589.68**

	List of Projects	QTY	Units	Cost Estimate
1	Grouhel Road	242	m	$ 142,700.00
2	Christie Road - Pictou Rd to Arroyo Rd	576	m	$ 339,700.00
3	Christie Road - East of Pictou Rd	738	m	$ 435,200.00
4	Harbour View & Inkster Rd	235	m	$ 138,600.00
5	Jones, Paton & Lipton Rd	639	m	$ 376,800.00

(rounded to nearest $100.00)

McElhanney Consulting Services Ltd.
2231-31902-01

DETAILED COST ESTIMATES

February 7, 2017

Project 4a – 150mm Diameter Watermain Replacement (Priority 6-30 years)

Description	QTY	Units	Rate	Extension	Total
150 mm dia. PVC Pipe	1960	m	$ 275.00	$ 539,000.00	
150mm gate valves	0	ea	$ 1,500.00	$ -	
hydrants	12	ea	$ 5,500.00	$ 66,000.00	
50mm asphalt cement, 1.2m wide	196	m	$ 42.00	$ 8,232.00	
75mm base gravel, 1.2m wide	196	m	$ 30.00	$ 5,880.00	
150mm subbase gravel, 1.2m wide	196	m	$ 72.00	$ 14,112.00	
shoulder gravel, 1.0m wide	1960	m	$ 25.00	$ 49,000.00	
standpipe	3	ea	$ 350.00	$ 1,050.00	
tie-in to existing watermain	10	ea	$ 7,500.00	$ 75,000.00	
				Subtotal	$ 758,274.00
				Contingency (30%)	$ 227,482.20
				Subtotal	$ 985,756.20
				Engineering (15%)	$ 147,863.43
				Total	**$ 1,133,619.63**
				Total Price/meter	**$ 578.38**

	List of Projects	QTY	Units	Cost Estimate
1	Christie Road, South of Grouhel	449	m	$ 264,800.00
2	Grouhel Road, Christie to Chandler	202	m	$ 119,100.00
3	Chandler Road	204	m	$ 120,300.00
4	Grouhel Road, Chandler to Pictou	649	m	$ 382,700.00
5	Pictou Road	354	m	$ 208,700.00
6	12556/12560 Island Hwy	102	m	$ 60,100.00

(rounded to nearest $100.00)

McElhanney Consulting Services Ltd.
2231-31902-01

DETAILED COST ESTIMATES

February 7, 2017

Project 4b – 150mm Diameter Watermain Replacement Under Highway (Priority 6-30 years)

Description	QTY	Units	Rate	Extension	Total
150 mm dia. HDPE	315	m	$ 325.00	$ 102,375.00	
Boring under Trans Canada Hwy	45	m	$ 375.00	$ 16,875.00	
150mm gate valves	2	ea	$ 1,500.00	$ 3,000.00	
hydrants	1	ea	$ 5,500.00	$ 5,500.00	
50mm asphalt cement, 1.2m wide	32	m	$ 42.00	$ 1,344.00	
75mm base gravel, 1.2m wide	32	m	$ 30.00	$ 960.00	
150mm subbase gravel, 1.2m wide	32	m	$ 72.00	$ 2,304.00	
shoulder gravel, 1.0m wide	158	m	$ 25.00	$ 3,937.50	
tie-in to existing watermain	2	ea	$ 7,500.00	$ 15,000.00	
				Subtotal	$ 151,295.50
				Contingency (30%)	$ 45,388.65
				Subtotal	$ 196,684.15
				Engineering (15%)	$ 29,502.62
				Total	**$ 226,186.77**
				Total Price/meter	**$ 718.05**

McElhanney D - 14-07-22

DIAMOND IMPROVEMENT DISTRICT

BC

EXISTING WATER SYSTEM

DEAD MAIN TO RESERVE (CLOSED) CAPPED OFF

EX. 100Ø - 290m

EX. 100Ø - 350m

EX. 100Ø - 500m

EX. 100Ø - 820m

EX. 100Ø - 240m

EX. 100Ø - 165m

EX. 100Ø - 70m

150Ø PVC

200Ø PVC

STANDPIPE VALVE

STANDPIPE

¾" FLUSHOUT

METER CHAMBER WITH PRESSURE REDUCING VALVES

VALVE 29.878 CONNECTION TO SUPPLY MAIN

LADYSMITH'S WATERMAIN

WV METAL LID 62.100

VALVE 62.130 CONNECTION TO SUPPLY MAIN

SHUTOFF

LIPTON ROAD

JONES ROAD

PATON ROAD

CHRISTIE ROAD

ARROYO ROAD

GROUHEL ROAD

PICTOU ROAD

TRANS CANADA HIGHWAY

ROCKY CREEK ROAD

INKSTER ROAD

HARBOUR VIEW ROAD

CHANDLER ROAD

COVENTRY LANE

LEGEND

EX. LENGTH 2,734m	EX. 150Ø OR 200 PVC WATERMAIN
EX. LENGTH 2,275m	EX. 150Ø AC WATERMAIN
EX. LENGTH 2,429m	EX. 100Ø AC WATERMAIN
	EX. WATER VALVE
	EX. FIRE HYDRANT

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0 1:2500 100

ORIGINAL DWG SIZE: (25" x 37")

McElhanney

McElhanney Consulting Services Ltd.

5855 York Road
Suite 202
Duncan BC
Canada V9L 3S3
Tel 250 748 6279

Approved Sealed

Drawing No. FIG 1

Project Number 31902-01

Rev. 0

DATE: 2017-04-21, 10:56 FILE: T:\Projects\2231 FILES\31902-01 Diamond Improvement District water system\10.0 Drawings\10.1 Active\2-Civil 3D\31902-01-DESIGN.dwg

DESTROY ALL PRINTS BEARING PREVIOUS REVISION

McElhanney D - 14-07-22

DEAD MAIN TO RESERVE (CLOSED) - CAPPED OFF

LIPTON ROAD

JONES ROAD

PATON ROAD

TRANS CANADA HIGHWAY

CHRISTIE ROAD

ROCKY CREEK ROAD

ARROYO ROAD

GROUHEL ROAD

PICTOU ROAD

INKSTER ROAD

HARBOUR VIEW ROAD

COVENTRY LANE

CHANDLER ROAD

EX. 100Ø - 280m

EX. 100Ø - 350m

EX. 100Ø - 500m

EX. 100Ø - 240m

EX. 100Ø - 820m

EX. 100Ø - 70m

EX. 100Ø - 165m

2000 PVC

VENT 7.193

VALVE 10.394

FIRE HYDRANT 27

kPa	psi
800	116
YEAR	2013

VALVE 4.870

FIRE HYDRANT 24

kPa	psi
841	122
YEAR	2013

VALVE 4.559

FIRE HYDRANT 25

kPa	psi
827	120
YEAR	2014

VALVE 9.660

FIRE HYDRANT 23

kPa	psi
814	118
YEAR	2013

VALVE 10.887

VALVE 9.176

FIRE HYDRANT 26

kPa	psi
800	116
YEAR	2013

VALVE 6.108

VENT 6.003

VALVE 12.190

VALVE 14.459

VALVE 19.176

FIRE HYDRANT 20

kPa	psi
772	112
YEAR	2013

VALVE 18.614

FIRE HYDRANT 22

kPa	psi
786	114
YEAR	2013

VALVE 17.490

VALVE 16.530

VALVE 17.234

FIRE HYDRANT 19

kPa	psi
786	114
YEAR	2013

FIRE HYDRANT 21

kPa	psi
758	110
YEAR	2013

VALVE 16.358

FIRE HYDRANT 18

kPa	psi
800	116
YEAR	2013

VALVE 14.967

VALVE 12.616

FIRE HYDRANT 17

kPa	psi
786	114
YEAR	2013

VALVE 12.683

VALVE 19.369

VALVE 19.603

FIRE HYDRANT 7

kPa	psi
662	96
YEAR	2014

VALVE 26.490

FIRE HYDRANT 28

kPa	psi
634	92
YEAR	2014

STANDPIPE VALVE

VENT 29.474

VALVE 25.890

FIRE HYDRANT 6

kPa	psi
662	96
YEAR	2014

VALVE 31.240

VALVE 18.766

FIRE HYDRANT 16

kPa	psi
724	105
YEAR	2013

3/4" FLUSHOUT

VALVE 27.665

VALVE 21.600

STANDPIPE

VALVE 13.990

FIRE HYDRANT 14

kPa	psi
621	90
YEAR	2013

FIRE HYDRANT 15

kPa	psi
662	96
YEAR	2013

VALVE 29.118

LADYSMITH'S WATERMAIN

VALVE 30.069

VALVE 30.334

VALVE 29.830

METER CHAMBER WITH PRESSURE REDUCING VALVES

VALVE 29.878 CONNECTION TO SUPPLY MAIN

VALVE 30.219

VALVE 36.550

FIRE HYDRANT 11

kPa	psi
565	82
YEAR	2014

LADYSMITH'S WATERMAIN

VALVE 39.198

FIRE HYDRANT 10

kPa	psi
524	76
YEAR	2014

VALVE 33.640

VALVE 32.060

FIRE HYDRANT 12

kPa	psi
579	84
YEAR	2014

VALVE 33.690

FIRE HYDRANT 13

kPa	psi
552	80
YEAR	2014

VALVE 36.649

FIRE HYDRANT 9

kPa	psi
538	78
YEAR	2014

VALVE 37.329

FIRE HYDRANT 8

kPa	psi
607	88
YEAR	2014

VALVE 39.290

VALVE 38.070

FIRE HYDRANT 1

kPa	psi
510	74
YEAR	2014

SHUTOFF

VALVE 37.290

VALVE 42.746

FIRE HYDRANT 3

kPa	psi
510	74
YEAR	2014

VALVE 45.970

FIRE HYDRANT 2

kPa	psi
455	66
YEAR	2014

VALVE 46.002

VALVE 46.170

FIRE HYDRANT 4

kPa	psi
421	61
YEAR	2014

VALVE 48.911

CONC LID 51.826

VENT & VALVE 51.781

WV METAL LID 62.100

VALVE 62.130 CONNECTION TO SUPPLY MAIN

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0 1:2500 100

ORIGINAL DWG SIZE: (25" x 37")

EX. 150Ø OR 200 PVC WATERMAIN
EX. 150Ø WATERMAIN
EX. 100Ø WATERMAIN
EX. WATER VALVE (SURVEY)
EX. WATER VALVE (NOT SURVEY)
EX. FIRE HYDRANT

McElhanney
McElhanney Consulting Services Ltd.
5855 York Road
Suite 202
Duncan BC
Canada V9L 3S3
Tel 250 748 6279

Approved Sealed

DIAMOND IMPROVEMENT DISTRICT

BC

EXISTING WATER SYSTEM FIRE HYDRANT FLOW DATA & VALVE ELEVATION

Drawing No. FIG 2

Project Number 31902-01

Rev. 0

DATE: 2017-04-21, 10:56 FILE: T:\Projects\2231 FILES\31902-01 Diamond Improvement District water system\10.0 Drawings\10.1 Active\2-Civil 3D\31902-01-DESIGN.dwg

DESTROY ALL PRINTS BEARING PREVIOUS REVISION